

CHEMISTRY GRADE 10: ACIDS AND BASES

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 3.0: Chemical Reactions and Mixtures
Sub-Strand	Sub-Strand 3.1: Acids and Bases
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Simple classification of substances as either acids or bases – Definition of acids and bases (Arrhenius Definition) – Chemical properties of acids and bases – Strength of acids and bases (pH scale, strong and weak acids and bases) – Uses of acids and bases
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) classify substances as either acids or bases using indicators in the laboratory and in everyday life b) define acids and bases using the Arrhenius definition c) investigate the chemical properties of acids and bases, including their reactions with metals, carbonates and each other (neutralisation) d) determine the strength of acids and bases using the pH scale and universal indicator, and distinguish strong from weak acids and bases e) appreciate the uses of acids and bases in day-to-day life, including safe handling and the use of antacids to relieve hyperacidity
Core Competencies	– Critical thinking and problem solving: learners interpret pH data for household products and hypothesise why each is acidic or basic – Communication and collaboration: learners discuss findings in groups, cold-call responses, and present models and explanations – Self-efficacy: learners carry out indicator tests and neutralisation investigations and gain confidence in laboratory work – Digital literacy: learners interpret graphs and use simulations of pH where available – Learning to learn: learners track their questions on the Driving Question Board and revise their models as evidence grows
Core Values	– Responsibility: handling acids, bases and household chemicals safely and disposing of them properly – Integrity: reporting observations honestly, even when they differ from predictions – Unity: working cooperatively in practical groups – Respect: caring for shared apparatus and for classmates' ideas
Science & Engineering Practices	– Asking questions and defining problems: learners generate questions about why products differ in pH and add them to the DQB

	– Analyzing and interpreting data: learners read the pH graph of household products and classify them – Planning and carrying out investigations: learners test local materials with indicators such as red cabbage extract, litmus and universal indicator – Developing and using models: learners build and revise particle models of H^+ and OH^- ions in solution – Constructing explanations: learners explain why antacids relieve hyperacidity using neutralisation – Engaging in argument from evidence: learners argue whether alkalinity determines a substance's use
Pertinent & Contemporary Issues (PCIs)	– Health education: safe use of antacids, avoiding misuse of household chemicals and understanding hyperacidity and ulcers – Safety and security: laboratory safety rules, hazard labels and storage of corrosive substances at home – Environmental education: effects of acidic and basic waste, acid rain and soil acidity on the environment – Life skills: making informed decisions on the use of cleaning and food products
Career Connections	– Pharmacist: formulates and dispenses antacids such as magnesium hydroxide and aluminium hydroxide – Agronomist/farmer: tests soil pH in Kericho tea farms and Rift Valley maize farms and applies lime to correct acidity – Food technologist: uses acids such as citric and ethanoic acid to preserve and flavour foods – Water and environmental analyst: monitors pH of Lake Victoria and industrial effluent – Medical doctor/clinical officer: treats hyperacidity, gastritis and ulcers
Focus for Lessons	This unit uses a graph of the pH values of everyday Kenyan household products to anchor learning about acids and bases. Learners predict, test, explain and model how the properties of acids and bases, their strength and their ionic nature determine their uses and safety precautions. Through investigations, learners build towards explaining why alkaline antacids are given to patients with hyperacidity.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications? KICD KEY INQUIRY QUESTIONS: 1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life?
Anchoring Phenomenon	The teacher displays a set of household products found in many Kenyan homes: lemon juice (from Kisumu market), baking soda (used for mandazi), bar soap, and vinegar. A graph shows the pH value of each product: lemon juice and vinegar have low pH values, while baking soda and soap have higher pH values above 7. Learners must interpret the data and hypothesise why each product is acidic or basic. It is puzzling because the products look similar (clear or white and found in the same kitchen), yet the lemon juice tastes sour and the soap feels slippery. Some are safe to eat, such as lemon juice and vinegar, while others, such as soap, are harmful if swallowed, and these properties influence how each product is used and what safety precautions are needed. Learners discuss how these properties relate to everyday uses, such as vinegar for pickling and soap for cleaning.
Storyline Thread	Lesson 1: Phenomenon – Students examine the pH graph of lemon juice, vinegar, baking soda and soap, predict why each is acidic or basic, and raise questions for the Driving Question Board. Lesson 2: Simple classification – Students test household substances with litmus and red cabbage extract and sort them as acids or bases. Lesson 3: Natural and laboratory indicators – Students compare indicator colour changes, build a classification table and revise their initial hypotheses. Lesson 4: Arrhenius definition (acids) – Students use models and simulations to show that acids produce H^+ ions in water and

	explain why lemon juice is sour. Lesson 5: Arrhenius definition (bases) – Students show that bases and alkalis produce OH^- ions in water, explain why soap feels slippery, and update their particle models. Lesson 6: Chemical properties I – Students investigate reactions of acids with metals and carbonates, such as vinegar with baking soda, and identify the gases produced. Lesson 7: Chemical properties II – Students investigate neutralisation between acids and bases and link it to how antacids relieve hyperacidity. Lesson 8: Strength of acids and bases – Students use universal indicator and the pH scale to compare strong and weak acids and bases, and relate strength to ion concentration. Lesson 9: Uses of acids and bases – Students analyse uses in food preservation, cleaning, farming (lime for acidic soil) and medicine, and evaluate safety precautions. Lesson 10: Consolidation – Students complete their final models, answer the driving question about alkalinity and antacids, and present claims using evidence from the unit.
Supporting Phenomena	– A person with heartburn after eating a heavy meal of nyama choma and fried foods takes an antacid tablet and feels relief – Tea farmers in Kericho add lime to acidic soil to improve crop yields – Lemon juice rubbed on a rusted or tarnished metal surface cleans it, but soap does not – Red cabbage extract changes colour when mixed with lemon juice, soap solution and baking soda

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (1 lesson)): Phenomenon: Why Do Lemon Juice, Vinegar, Baking Soda and Soap Behave So Differently?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by confronting learners with the puzzling phenomenon that four similar-looking household products differ sharply in taste, feel, pH and safety, and to open the Driving Question Board (DQB).
Knowledge	Learners interpret the pH graph to see that lemon juice (pH 2.0) and vinegar (pH 2.8) fall below 7, while baking soda solution (pH 8.3) and soap solution (pH 10.0) fall above 7, with pure water (pH 7.0) as the reference. They link sour and slippery observations to these values and to how the products are used.
Skills	Interpreting data from a graph (SEP 4: Analysing and Interpreting Data); asking questions (SEP 1); making predictions and generating hypotheses; communicating ideas in pairs and as a class.
Attitudes	Curiosity about everyday substances, respect for peers' ideas, and a safety-conscious attitude: never taste or touch unknown chemicals in the laboratory or at home.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life?
Purpose in Storyline	Lesson 1 launches the phenomenon and the DQB. Learners notice that the products fall on either side of 7 and propose initial ideas and questions. These ideas are revisited in Lessons 2-3, where learners classify the same products using indicators, and they seed the particle model that is revised in every later lesson. Learners are not told WHY the products differ; that is built across the unit.
Safety Notes	The products are displayed only for looking. Nobody tastes lab chemicals. Learners must not taste the products in class; sour and slippery observations are given as reported data (lemon juice and vinegar are safe in food, baking soda is safe in small amounts in food, and soap is harmful if swallowed). Learners may touch the soap solution only if the teacher allows it, then wash their hands. Use clean, labelled containers and keep soap solution away from eyes.

B. LESSON OVERVIEW

Mrs Achieng's class in Kisumu sees four products on the front bench: lemon juice bought from Kisumu market, vinegar used by mama mboga for pickling kachumbari, baking soda used by Mama Wanjiru for mandazi, and a bar of soap. In clear cups they all look similar. Yet Otieno asks why lemon is sour, and Wanjiru asks why soap is slippery. Learners predict how the products differ, study a pH graph (lemon juice 2.0, vinegar 2.8, pure water 7.0, baking soda 8.3, soap 10.0), build hypotheses, and raise questions for the DQB. The driving question is: Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications? They draw an initial particle model, which will be revised in every lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Definition of pH Source: Khan	Learners look at the four products displayed in clear cups on the teacher's bench: lemon juice, vinegar, baking soda solution and	Display the products before learners enter, with labels. Say: 'These four things sit in many Kenyan kitchens. Look at them	Predict-Write-Share: writing a prediction before instruction makes learners' prior ideas visible and gives them something to compare	Check whether learners (1) give a reason (taste, feel, use) rather than only a guess, (2) distinguish edible items from harmful ones,

Academy (English - US curriculum) Search ARES for similar videos PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content Search ARES for similar readings	soap solution. In pairs (3 minutes) they answer in their notebooks: (a) Which products do you think are safe to eat and which are harmful? (b) Which are alike and which are different? (c) Predict how the pH values would be arranged on a scale from 0 to 14 for these products. They write a quick prediction and a reason, using what they know from the kitchen at home, for example lemon in tea, vinegar on kachumbari, and soap for washing clothes.	carefully. Do not taste or touch them.' Ask: 'Without any equipment, how could we tell them apart?' WAIT TIME 10-15 seconds before taking answers. Cold-call 4 learners for their predictions and record them on the board under 'Our first ideas' without judging as right or wrong. Then ask: 'Otieno says lemon juice is sour and Wanjiru says soap is slippery. Why might that be?' WAIT TIME 10-15 seconds. Tell learners we will test these ideas with data.	with the evidence later. Recording ideas without judging encourages risk-taking and creates a baseline to revisit when the pH data arrive.	and (3) name properties such as sour, slippery or bitter. Note in your checklist which learners suggest a number-based or scale-based way to compare. Misconceptions to note: 'all chemicals are dangerous', 'soap is the same as lemon because both clean', or 'clear liquids are the same'.
Observe Phase VIDEO: Definition of pH Source: Khan Academy (English - US curriculum) Search ARES for similar videos PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content Search ARES for similar readings	Learners study the pH graph, a bar graph of pH values measured with a pH meter at 25 °C: lemon juice (Kisumu market) 2.0, vinegar (5% ethanoic acid, shop-bought) 2.8, pure water 7.0 (reference line), baking soda solution (1 spoon in 1 cup water, as used for mandazi) 8.3, bar soap solution (1 spoon shavings in 1 cup water) 10.0. Using a data table, they (1) put the products in order from lowest to highest pH, (2) mark which are below and above the water line at 7, (3) calculate differences, such as lemon juice and vinegar differing by 0.8 and soap and baking soda by 1.7, and (4) match each product with the observations given by the teacher: lemon juice and vinegar taste sour and are safe in food; baking soda tastes bitter/salty and is safe in small amounts in food; soap feels slippery, tastes bitter and is harmful if swallowed. They also note the uses: vinegar for pickling, soap for cleaning, baking soda for raising mandazi, lemon juice for flavour and tea.	Give each pair a printed graph and observation card. Say: 'Look at the water line at 7. What do you notice about where each product sits?' WAIT TIME 10-15 seconds. Cold-call 3 pairs for their observation, then 2 more for a different pattern. Prompt: 'Which two products are close together? Which is furthest from water?' Ask: 'Lemon juice is 2.0 and vinegar is 2.8. What is the difference, and does it matter?' WAIT TIME 10-15 seconds. Do NOT explain H⁺ or OH⁻ ions or the 10-fold rule; if asked, say: 'Great question, put it on our Driving Question Board.' Read out the sour and slippery observations clearly and stress that nobody should taste lab chemicals. Circulate and check that learners read the axis correctly.	Data-pattern noticing (SEP 4): a structured 'I notice / I wonder' routine, with the graph as the only source of evidence. Ordering products and marking the line at 7 helps learners see that the products fall on both sides of pure water, which is the core idea for classification in later lessons.	Look for learners who (1) correctly order the products as lemon juice (2.0) < vinegar (2.8) < water (7.0) < baking soda (8.3) < soap (10.0), (2) state that two products are below 7 and two are above 7, (3) calculate differences accurately, and (4) link sour with low values and slippery with high values as a pattern, not yet as an explanation. Common errors: reading the graph from the wrong axis, thinking a higher number means more acidic, or ignoring the water reference.
Explain Phase VIDEO:	In groups of four, learners build a hypothesis for each product using	Say: 'Work as a team of scientists. Your hypothesis must use at least	Claim-Evidence-Reasoning with an Initial Model (SEP 2:	Check whether hypotheses (1) cite numbers from the graph, (2)

Definition of pH Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Strong and Weak Bases and Base Ionization Constant (Kb) (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	a claim-evidence-reasoning frame: 'We think lemon juice is ____ because the graph shows ____ and it tastes ____.' They discuss: Why are lemon juice and vinegar so different from soap, although all look clear or white? Why is vinegar good for pickling kachumbari while soap is good for cleaning? Why can soap be harmful if swallowed while lemon juice is safe? They propose hypotheses about what might be inside each product that makes it behave this way. They then sketch an INITIAL MODEL: four boxes (lemon juice, vinegar, baking soda, soap) showing particles they think are responsible for sourness or slipperiness, using their own symbols and a key. Wrong ideas are welcome at this stage.	one number from the graph.' Circulate and ask: 'What in the data makes you say that?' and 'What might be different inside the lemon juice compared with the soap?' WAIT TIME 10-15 seconds. Cold-call 4 groups to share one hypothesis each; after each, ask a different group: 'Do you agree or disagree, and what evidence would help?' Introduce the words 'acidic' for values below 7 and 'basic' for values above 7 as labels for the pattern only, and say that 7 is neutral like pure water; do not explain ions. Tell groups: 'Draw what you think is inside, even if you are not sure. We will fix it as we learn.' Collect two or three different models and display them.	Developing and Using Models): turning the pattern into a hypothesis and a drawing makes learners' thinking visible and creates a baseline model that is revised in every later lesson. Encouraging unfinished or wrong models makes later revisions meaningful.	connect pH to taste and feel, and (3) suggest something inside the substances (particles, ingredients, e.g. citric acid in lemon juice, fats and sodium hydroxide in soap). Look for models that show different particles for different products. Note misconceptions such as 'soap is slippery because of oil only', 'acids are always dangerous', or 'neutral means harmless'. Do not correct; record them for later lessons.
Driving Question Board (DQB) Creation VIDEO: Definition of pH Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Strong and Weak Bases Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners read the driving question: 'Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications?' They write 2 questions each on sticky notes, for example 'Why is lemon sour?' (Otieno), 'Why is soap slippery?' (Wanjiru), 'How is pH measured without a meter?', 'What is in an antacid that helps after eating nyama choma?' In groups they sort the notes into categories such as 'How to tell acid from base', 'What is inside them', 'Uses and safety', and 'Health and medicine', and stick them on the DQB under the driving question. Each group reads out its best question.	Introduce the board: 'This is our Driving Question Board. It tracks what we wonder and what we learn.' Prompt: 'Think of a nurse in Nairobi giving an antacid to a patient with hyperacidity after eating spicy nyama choma. What would you want to know?' WAIT TIME 10-15 seconds. Cold-call 5 learners to read a question aloud and place it. Ask the class: 'Which questions can we answer by testing? Which need us to learn what is inside the particles?' Link to the KICD key inquiry questions by writing on the board: 'How do we classify substances as acids or bases?' and 'Why are acids and bases important in our day-to-day life?' Tell learners we will return to the board at the end of each lesson, move questions to 'answered', and add new ones. Do not answer any question today.	Question-Sorting Routine (SEP 1: Asking Questions): sorting learner questions into categories gives purpose to the learning sequence and shows learners that their curiosity drives the lessons. Linking questions to investigable ideas builds a shared 'need to know'.	Count questions per learner. Look for questions that (1) ask 'why' or 'how' rather than only 'what', (2) relate to the phenomenon (taste, feel, pH, safety), and (3) connect to the driving question (uses, antacids, hyperacidity). Note learners who only copy others and prompt them with: 'What did you see on the graph that surprised you?'

Model Building Phase  VIDEO: Definition of pH Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Identifying weak bases and strong bases <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revisit their initial model and complete a 'Summary Table' in their notebooks with three columns: What we observed (e.g. lemon juice 2.0 and vinegar 2.8 are below 7; baking soda 8.3 and soap 10.0 are above 7; sour versus slippery), What we learned (the products fall on both sides of pure water at 7, and uses differ), and What we can explain so far (a tentative hypothesis). They add one sentence: 'I think hyperacidity patients get antacids because ____.' They finish by writing a prediction for Lesson 2: 'How could we tell if a substance is an acid or a base without a pH meter?'	Say: 'Open your model page. Label which products you think are on the acidic side and which are on the basic side.' WAIT TIME 10-15 seconds. Cold-call 3 learners to read their 'I think hyperacidity patients get antacids because...' statements; accept all and say: 'We will test this idea with evidence.' Ask: 'What is one thing you are still unsure about?' WAIT TIME 10-15 seconds. Collect models for comparison later. Preview: 'Next lesson we will classify these same four products using colour changes of litmus and red cabbage.' Remind learners about safety: never taste unknown substances at home or in the lab.	Summary Table and Model Baseline (SEP 2): organising observations, learning and tentative explanations into a table builds a record that learners can compare with their later understanding. A model kept as a baseline reveals the progress of their ideas across the unit.	Collect notebooks and check that (1) the observed column includes accurate pH values and the pattern that two products are below 7 and two are above, (2) the learned column states that products fall on both sides of 7, (3) the explained column holds a tentative idea, even if incomplete, and (4) the model has a key. Use a quick tally: how many learners can place all four products on the correct side of 7? Learners who cannot get extra practice with the graph next lesson.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners notice the pattern around 7 without being told? Which hypotheses about what is inside the products were most common, and which misconceptions (e.g. all acids are dangerous, soap is just oil, clear means the same) will need to be addressed in Lessons 2-5? Did the Kenyan contexts (Kisumu market lemons, mandazi, kachumbari, nyama choma) make the phenomenon feel real? Which learners asked the strongest DQB questions, and which stayed silent? Was WAIT TIME of 10-15 seconds used consistently? What will you adjust for Lesson 2, where learners classify the same four products with litmus and red cabbage extract?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that lemon juice (pH 2.0) and vinegar (pH 2.8) have pH values below pure water (pH 7.0), while baking soda solution (pH 8.3) and soap solution (pH 10.0) are above 7. Lemon juice and vinegar taste sour and are safe in food, baking soda tastes bitter/salty and is safe in small amounts in food, and soap feels slippery and is harmful if swallowed.
What did I learn?	We learned that products that look alike can be on different sides of 7 on the pH scale, and that their uses (vinegar for pickling, soap for cleaning, baking soda for mandazi, lemon juice for flavour and tea) seem to match their pH, taste and feel.
How does this explain the phenomenon?	We can so far hypothesise that something different inside lemon juice and vinegar from inside baking soda and soap causes the difference in pH, taste and feel, and that antacids may help hyperacidity because they are on the other side of the scale from stomach acid. We still need evidence to test this idea.

LESSON 2 (80 minutes (2 x 40-minute periods)): Sorting the Kitchen: Classifying Household Substances as Acids or Bases with Litmus and Red Cabbage

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners classify lemon juice, vinegar, baking soda solution and bar soap solution as acids or bases using the colour changes of litmus and red cabbage extract, and use this evidence to check the hypotheses they made in Lesson 1.
Knowledge	Acids turn blue litmus red and leave red litmus red; bases turn red litmus blue and leave blue litmus blue. Red cabbage extract (purple) turns red/pink in acids, stays purple in water, and turns blue-green to green-yellow in bases. Indicators are substances that show by colour change whether a substance is acidic or basic. Classification must be based on evidence from a pattern of colour changes, not on appearance alone.
Skills	Planning and carrying out a fair test (NGSS SEP: Planning and Carrying Out Investigations); recording observations in a results table; analysing and interpreting data to identify patterns (NGSS SEP: Analyzing and Interpreting Data); constructing evidence-based claims (NGSS SEP: Engaging in Argument from Evidence); handling chemicals safely.
Attitudes	Curiosity about how everyday kitchen substances behave; honesty in recording what is actually observed; respect for safety rules; willingness to revise a hypothesis when evidence disagrees; cooperation in groups.
Key Inquiry Question	How do we classify substances as acids or bases? (KICD Key Inquiry Question 1) Linked to: Why are acids and bases important in our day-to-day life?
Purpose in Storyline	In Lesson 1 learners saw the pH graph for lemon juice, vinegar, baking soda and soap and suggested hypotheses about which are acidic or basic. In this lesson they gather their first hands-on evidence: indicator colour changes that sort the four products into two groups. This gives them a classification they can later explain with H ⁺ and OH ⁻ ions (Lessons 4-5) and test with other indicators (Lesson 3), and moves them towards answering whether the nature of a substance decides its use.
Safety Notes	Wear goggles and wear aprons. NEVER taste any substance in the laboratory, including the kitchen products, because lab containers and equipment are not food-safe. Use only small volumes (about 2 cm depth in test tubes or small cups). Soap solution may irritate eyes; rinse splashes with plenty of water and tell the teacher. Wash hands after the practical. Dispose of used solutions down the sink with plenty of water. Prepare the red cabbage extract beforehand with warm water, and the teacher handles all hot water.

B. LESSON OVERVIEW

Mrs Achieng's Kisumu class return to the four products from Lesson 1 (lemon juice from Kisumu market, vinegar, baking soda used by Mama Wanjiru for mandazi, and bar soap), which look alike but behave differently. Learners first predict how litmus paper and purple red cabbage extract will change in each product, using their Lesson 1 hypotheses. They then test each product (plus pure water as a reference) with red litmus, blue litmus and red cabbage extract, and record their results. In the Explain phase they look for patterns in the class data, sort the products into acids and bases, and compare the sorting with their Lesson 1 hypotheses. They add new questions to the Driving Question Board and add a first classification layer to their model (labelling each product as acid or base, with particle drawings of H⁺ and OH⁻ coming in later lessons).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Conjugate acid–base pairs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	Learners look at the four products again (lemon juice, vinegar, baking soda solution, soap solution) and the pH graph from Lesson 1. Otieno's question 'Why is lemon sour?' and Wanjiru's question 'Why is soap slippery?' are read out. Individually, learners write in their notebooks a prediction for each product: will red litmus turn blue, will blue litmus turn red, and what colour will purple red cabbage extract become (red/pink, stay purple, or blue/green)? They write one sentence explaining the reason for their prediction, using ideas from Lesson 1. Then they compare with a partner.	Display the four products and the pH graph. Say: 'Last lesson you made hypotheses about which of these are acidic and which are basic. Today we will collect evidence to check them. A hypothesis is only useful if we can test it.' Show red litmus, blue litmus and a cup of purple red cabbage extract and say: 'Before we test, write down what you think will happen to each of these in each product, and why.' Give WAIT TIME of 10-15 seconds of silent thinking before writing begins. Circulate and note common ideas, for example that appearance (clear vs white) decides acidity. Cold-call 3 learners with: 'What do you predict for the soap and why?' Ask 2 more: 'Who predicted something different?' Record the predictions on the board in a prediction table without saying which is correct.	Prediction with Reasoning (Predict-Observe-Explain): Writing a prediction and a reason before the experiment makes learners' existing ideas visible. When their results later agree or disagree with the prediction, they have a clear reason to revise their thinking, which builds conceptual change.	Check notebooks for: (1) a prediction written for each of the four products; (2) a reason that refers to Lesson 1 evidence (pH below or above 7, sour taste or slippery feel) and not only appearance. Indicator of readiness: at least 80% of learners link low pH or sour taste to 'acid' and higher pH or slippery feel to 'base'. Note learners who predict from appearance alone for extra questioning during the practical.
Observe Phase  VIDEO: Acid–base properties of salts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES	In groups of four, learners carry out the practical. Roles: Materials manager, Tester, Recorder and Safety/Reporter. They label 5 small cups: lemon juice, vinegar, water, baking soda solution (1 spoon in 1 cup water) and soap solution (1 spoon shavings in 1 cup water). Step 1: Dip a strip of red litmus into each, record the colour. Step 2: Dip a strip of blue litmus into each, record the colour (using a fresh strip each time). Step 3: Add about 10 drops of purple red cabbage extract to about 2 cm of each liquid and record the colour. They record in a results table with columns for substance, red litmus, blue litmus and red cabbage colour. Expected	Before starting, demonstrate the safety rules: 'Goggles on. We never taste anything in the lab, even if it is food at home.' Model how to use a fresh strip each time and how to avoid cross-contamination. Say: 'Water is our reference. Why do you think we test water too?' and give WAIT TIME of 10-15 seconds. Circulate with questions: 'What colour did the red litmus become in the soap? What does that tell you?' and 'Is this what you predicted?' Challenge: 'Why must we use the same amount of each liquid and the same number of drops?' (fair test). Cold-call 3 groups to report one surprising observation. Check any group whose cabbage colours	Structured Data Collection with a Fair Test: A shared results table and a water control make the evidence systematic and comparable. Learners see the same pattern repeating across three indicators, which supports reliability and starts pattern recognition.	Observe whether: (1) each group uses a fresh strip for each liquid; (2) results tables are complete and colours are described accurately (for example 'blue-green', not just 'blue'); (3) water is used as a reference; (4) learners wear goggles and do not taste anything. Indicator of success: at least 4 of 5 rows in a group table match the class pattern. Use a quick checklist and prompt groups with errors to repeat.

for similar readings	results: red litmus turns blue in baking soda and soap but stays red in lemon juice, vinegar and (for water) unchanged; blue litmus turns red in lemon juice and vinegar but stays blue in baking soda and soap; red cabbage extract is red/pink in lemon juice, pink-red in vinegar, purple in water, blue-green in baking soda and green-yellow in soap. They also compare with their predictions and mark each one correct or incorrect.	do not match the expected pattern and ask them to repeat with fresh liquid. Remind groups to record exactly what they see, not what they expect.		
Explain Phase  VIDEO: Conjugate acid–base pairs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	Groups put their results on a large class table on the board (one column per indicator, one row per product). In pairs, learners look for patterns: which products turn red litmus blue and which turn blue litmus red, and how the red cabbage colours group together. They then write a claim: 'Lemon juice and vinegar are ____ because ____; baking soda and soap are ____ because ____.' They use evidence from at least two indicators. They compare the classification with their Lesson 1 hypotheses and the pH graph (lemon 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0): did the indicators agree with the pH side of 7? They explain why water is neither (neutral) and discuss why the colour of the cabbage extract is a range rather than just two colours. They then relate the finding to Otieno's and Wanjiru's questions, noting that they know what group each belongs to but not yet why, which is saved for later lessons. They also discuss, with home examples, what the sorting suggests about uses (for example vinegar for pickling kachumbari and onions, soap for	Ask: 'What pattern do you see across the three indicators? Talk to your partner for two minutes.' Use WAIT TIME of 10-15 seconds after each question. Cold-call 4 learners to describe the pattern using evidence. Prompt: 'Which hypothesis from last lesson was supported? Which needs revising?' Use the sentence frame: 'I know lemon juice is an acid because the blue litmus turned ____ and the cabbage extract turned ____.' Challenge misconceptions, for example 'soap is a base because it is not food' or 'white means base': 'Does colour or appearance appear in our evidence? What does our evidence say?' Ask: 'Water stayed purple. Where would you place it?' Introduce the working definitions: 'An indicator is a substance that changes colour to show whether something is acidic or basic. Acids turn blue litmus red; bases turn red litmus blue.' Note that alkali and ion explanations will come later. Do not give the ion explanation yet.	Claim-Evidence-Reasoning (CER) with Pattern Analysis: Learners state a claim, cite colour-change data from more than one indicator, and give reasoning linking the pattern to the acid/base category. Using multiple indicators strengthens the argument and shows why classification rests on evidence.	Collect written CER statements. Success criteria: (1) a correct claim for all four products; (2) evidence cited from at least two indicators with correct colours; (3) a reason connecting the colour pattern to acid or base; (4) water identified as neutral. Look for learners who still rely on taste or looks, and plan re-teaching with the data table. Use a thumbs-up/sideways/down check on 'I can classify a substance using litmus'.

	cleaning, baking soda for mandazi).			
Driving Question Board (DQB) Creation  VIDEO: Acid-base properties of salts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	Learners revisit the Driving Question Board from Lesson 1: 'Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications?' In groups they write new questions on sticky notes arising from today's evidence, for example 'Why does litmus change colour?', 'Why does red cabbage show many colours?', 'Why does a kitchen product like baking soda behave like soap?', 'How strong is each acid?' and 'Is an antacid an acid or a base?' They place their notes on the DQB under categories (How do we know?, Why does it happen?, What is it used for?) and mark which earlier questions have been partly answered (for example 'Which products are acidic and which basic?').	Say: 'Look at the board. Which of our questions did we make progress on today? Which new questions do you have?' Give WAIT TIME of 10-15 seconds, then ask groups to write at least 2 new questions each. Cold-call 3 groups to read a question and say where it goes on the DQB. Highlight a question the class cannot yet answer, such as 'Why does soap feel slippery?', and say: 'That is a great question. We will return to it when we study what is inside these substances.' Tick off or annotate answered questions with a short note, for example 'Lesson 2: classified using litmus and cabbage'.	Driving Question Board (DQB): Placing new questions publicly connects each lesson's evidence to the unit's big question. It shows learners that their curiosity guides what is investigated next and keeps the phenomenon at the centre.	Check that each group has produced at least 2 questions that arise from the evidence (not general topic questions). Indicator of progress: questions about causes ('why does the colour change?') or uses ('why use baking soda in mandazi?'), not only recall. Note learners who ask about antacids as an early link to the second part of the driving question.
Model Building Phase  VIDEO: Conjugate acid-base pairs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content	Learners open the initial model page from Lesson 1 and begin the first version of the class model. Using four boxes (lemon juice, vinegar, baking soda solution, soap solution) plus water, they label each box 'acid' or 'base' (and water 'neutral'), add the colour results for litmus and red cabbage, and write the pH value from the graph (2.0, 2.8, 8.3, 10.0, and 7.0). Where particles are not yet known, they leave a labelled 'we do not know yet' space for what is inside that makes the substance act this way, and draw a question mark. They add a short caption linking one product to its use, such as 'vinegar (acid) is used by mama mboga to pickle kachumbari' or	Say: 'Our model should show what we have evidence for. What do we know, and what do we still not know?' Provide WAIT TIME of 10-15 seconds before drawing. Circulate and ask: 'Which evidence supports the label on this box?' and 'What could be different inside lemon juice and soap?' Cold-call 3 learners to share a label and the evidence behind it. Emphasise that the particle drawings with H⁺ and OH⁻ will be added in later lessons. Collect model pages to review.	Model Building and Revision: Adding evidence-based labels and honest gaps to a visible model helps learners link observations to a developing explanation, and shows that models grow as new evidence is gathered.	Check models for: (1) correct labels for all five substances; (2) colour results and pH values placed correctly; (3) a clearly marked unknown area instead of invented explanations; (4) a use linked to the classification. Exit ticket: 'Name one piece of evidence that shows lemon juice is an acid and one that shows soap is a base.' Success is two correct pieces of evidence.

 Search ARES for similar readings	'baking soda (base) is used for mandazi'. They revise any hypothesis that was not supported.			
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D. TEACHER REFLECTION				
Which learners relied on appearance or taste instead of the indicator evidence, and how did their thinking change after the practical? Did every group use a fair test with a water reference and fresh strips? Were the red cabbage colours distinct enough for learners to see the difference between lemon juice, vinegar and water? Which questions on the DQB show that learners are ready to ask why indicators change colour, and how can I use them in Lessons 3 to 5? Did the classification activity stay linked to the phenomenon and to the idea of use (pickling, mandazi, cleaning)?				

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Record what you saw: the colour of red litmus, blue litmus and red cabbage extract in lemon juice, vinegar, water, baking soda solution and soap solution.
What did I learn?	Write how you can sort a substance as an acid or a base using the pattern of colour changes: acids turn blue litmus red; bases turn red litmus blue; water does not change either.
How does this explain the phenomenon?	Explain how the indicator results classify the four kitchen products, how they compare with your Lesson 1 hypotheses and pH values (2.0, 2.8, 7.0, 8.3, 10.0), and what you still want to find out about why they behave this way and how this might affect their uses.

LESSON 3 (80 minutes (2 x 40-minute periods)): Natural and Laboratory Indicators: Building a Classification Table

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners compare natural and laboratory indicators using the four household products and water, and build a classification table (acid, neutral, base) that lets them check and revise their Lesson 1 hypotheses against the pH values.
Knowledge	Acids are sour; bases are slippery and bitter. Litmus is red in acids and blue in bases. Phenolphthalein is colourless in acids and pink in bases. Methyl orange is red in acids and yellow in bases. Red cabbage extract is a natural indicator (purple in water, red/pink in acids, blue-green to green-yellow in bases). Universal indicator, a laboratory indicator, gives a range of colours that matches the pH scale (0 to 14; 7 neutral; below 7 acidic; above 7 basic). Pure water is neutral (pH 7.0).
Skills	Observing and recording colour changes accurately; comparing the evidence from different indicators; constructing a classification table; analysing data (matching colours to pH values); revising hypotheses using evidence; communicating findings in scientific language (NGSS SEP: Analyzing and Interpreting Data; Constructing Explanations).
Attitudes	Curiosity about everyday Kenyan products; respect for evidence when it contradicts an earlier idea; careful and responsible handling of chemicals; cooperation and respect for peers' ideas in group work.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life?
Purpose in Storyline	Lesson 2 gave learners a first sort of lemon juice, vinegar, baking soda and soap using litmus and red cabbage. Lesson 3 adds universal indicator (and optional phenolphthalein), puts neutral water into the sort, and links colour to pH values so learners can test the hypotheses from Lesson 1. This gives the classification evidence that Lesson 4 will explain with H ⁺ ions.
Safety Notes	Wear goggles and keep food items out of the lab; never taste any chemical or test solution, including the household products used as test solutions. Use only the household products at the dilutions given (1 spoon in 1 cup water for baking soda and soap). Phenolphthalein, if used, is for the teacher demonstration only. Wipe spills immediately (soap is slippery), wash hands after the practical, and dispose of the test solutions down the sink with plenty of water.

B. LESSON OVERVIEW

Mrs Achieng returns to the Kisumu market products and the pH graph. Otieno asked 'why is lemon sour?' and Wanjiru asked 'why is soap slippery?', and the class has so far sorted the products with litmus and red cabbage. Today learners predict which colours universal indicator will give, then test lemon juice, vinegar, water, baking soda solution and soap solution with red cabbage extract and universal indicator, and compare the two indicators. Learners explain why different indicators give different colours but the same sorting, build a three-column classification table (acid, neutral, base) that includes water, match colours to the pH values (2.0, 2.8, 7.0, 8.3, 10.0), revise their hypotheses and the Driving Question Board (DQB), and update their particle model with a first claim about which products are likely to have more H⁺ and which more OH⁻.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the Lesson 2 litmus	Display the four products and a	Predict-Observe-Explain	Look for: (1) at least 8 of 10 pairs

 VIDEO: 03 Natural Indicators Source: Seavuria  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	and red cabbage sorting (acidic vs basic). Mrs Achieng shows the pH graph and asks: 'Lemon juice is pH 2.0 and vinegar is 2.8. Both are acids, but will they look the same with universal indicator? Where will pure water fit?' In pairs, learners draw a prediction strip of 5 colour boxes (lemon juice, vinegar, water, baking soda, soap) and colour it using a rainbow guess, with a sentence: 'I think ____ will be ____ because ____.' Learners also predict whether water belongs in the acid column, the base column, or neither.	glass of water from the Kisumu school tap. Say: 'Yesterday we sorted by litmus. Today water joins the sorting. Where does it go, and how could we tell?' Give WAIT TIME of 10-15 seconds before anyone speaks. Cold-call 4 pairs (not volunteers) to share the colour they predict for lemon juice and why. Record all predictions on the board without judging. Ask: 'Which of the 5 colours do you feel most sure about? Which least sure?' Note learners who say water is acidic or basic, and learners who link low pH to red.	(prediction strip): committing to a prediction makes learners' existing ideas visible, so later evidence either confirms or challenges them. This creates a reason to look carefully at the results and fits the idea of testing hypotheses from Lesson 1.	commit to a written prediction for all 5 substances; (2) learners use evidence from Lesson 2 (e.g. 'lemon turned blue litmus red so it should be at the red end'); (3) learners place water in a separate or neutral position. Note misconceptions such as 'water has no pH' or 'all acids are the same colour'.
Observe Phase  VIDEO: 03 Natural Indicators Source: Seavuria  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	In groups of 4, learners use 5 labelled test tubes (lemon juice, vinegar, water, baking soda solution, soap solution; as in Lesson 1, the baking soda and soap are 1 spoon in 1 cup water). Step 1: add 2 cm³ of red cabbage extract (purple) to each tube and record the colours. Step 2: in a second set of tubes add 3 drops of universal indicator and compare with the pH colour chart. Expected results: Red cabbage extract: lemon juice red/pink, vinegar pink-red, water purple, baking soda blue-green, soap green-yellow. Universal indicator: lemon juice red-orange (pH 2), vinegar orange (pH 3), water green (7), baking soda blue (8), soap blue-purple (10). The teacher demonstrates phenolphthalein (optional): colourless in lemon juice and vinegar, colourless in water, pink in baking soda and soap solutions. Groups record in a table: substance red cabbage colour universal indicator colour pH from chart.	Model one tube first: 'Watch carefully as I add three drops to the lemon juice. Describe only what you SEE, not what you think.' Circulate with a checklist; ask each group: 'Which two tubes look most alike with red cabbage? With universal indicator? What does that tell you?' Remind: 'Goggles on. We do not taste, even the lemon juice and vinegar.' Wait 10-15 seconds after asking: 'What do you notice about the order of colours from lemon juice to soap?' Cold-call 3 groups to read out their colour for water. Point to the pH graph to show that the chart readings (pH 2, 3, 7, 8, 10) are close to the pH meter values (2.0, 2.8, 7.0, 8.3, 10.0), and explain that the chart is read by colour, so it gives whole numbers while the meter gives more exact values.	Structured data collection with a shared observation table: separating observation (colour seen) from interpretation (acid or base) trains learners to use evidence rather than assumptions, and comparing two indicators on the same samples shows what each indicator adds.	Look for: (1) groups record colours for all 5 substances with both indicators; (2) learners use descriptive colour terms (red-orange, blue-green) rather than 'it changed'; (3) learners notice that water is purple with red cabbage and green with universal indicator, unlike the other four; (4) learners safely handle tubes and keep goggles on. Ask one learner per group to point to the tube that gave the lowest pH reading.
Explain Phase	Learners build a class	Draw the empty table and ask	Claim-Evidence-Reasoning with a	Look for: (1) the table has water

 VIDEO: Acid-base properties of salts <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	classification table with three columns: Acid (pH below 7), Neutral (pH 7), Base (pH above 7). Under each substance they list: indicator colours (litmus from Lesson 2, red cabbage, universal indicator), pH value (from the graph), taste/feel observation (sour, bitter, slippery) and everyday use (vinegar for pickling kachumbari, soap for cleaning, baking soda for mandazi, lemon for tea). Learners then compare indicators: red cabbage extract is a natural indicator made from a plant; universal indicator is a laboratory indicator; litmus is a laboratory indicator (from lichen). Groups answer: (a) Why is the sorting the same with all indicators but the colours differ? (b) Which indicator tells us how acidic or basic, not just which side? (c) Which of our Lesson 1 hypotheses were supported or need revising? Learners write a claim-evidence-reasoning statement, e.g. 'Lemon juice is more acidic than vinegar because universal indicator is red-orange (pH 2) while vinegar is orange (pH 3), and the graph shows 2.0 and 2.8.'	groups to propose entries; say: 'Where does water go and what is your evidence?' WAIT TIME 10-15 seconds. Cold-call 5 learners, one for each substance, to place it and give the evidence. Probe: 'Litmus only told us acid or base. What extra information did universal indicator give us?' Ask: 'Otieno asked why lemon is sour. Can the table answer that yet?' (Expected: not yet; the table shows that it is acidic, not why it is sour. Park for the DQB.) Ask learners to compare their Lesson 1 hypotheses with the table: 'Which hypothesis do you keep, and which do you change?' Clarify the convention: acid produces H^+ (aq) in water; do NOT explain ions in depth, as that is for Lesson 4. Note that red cabbage is an everyday natural indicator that can be extracted from the kitchen, whereas litmus and universal indicator come from the lab.	classification table: organising multiple lines of evidence in one table lets learners see patterns (low pH = red/orange = acid; high pH = blue/purple = base; water in the middle) and justify classifications with data. Comparing hypotheses with evidence builds the habit of revising ideas.	correctly in the neutral column with pH 7.0 and a green universal indicator colour; (2) learners cite at least 2 pieces of evidence per classification; (3) learners state that lemon juice (pH 2.0) is more acidic than vinegar (pH 2.8), and soap (10.0) more basic than baking soda (8.3); (4) learners revise at least one hypothesis in writing. Use an exit slip: 'Name one advantage of universal indicator over litmus.' Expected: it shows the strength (pH value), not only acid or base.
Driving Question Board (DQB) Creation  VIDEO: Titration curves and acid-base indicators <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	At the Driving Question Board, learners add sticky notes in two colours: green = 'What we now know' and yellow = 'New questions'. Possible green notes: 'Water is neutral, pH 7.0', 'Lemon juice has a lower pH than vinegar', 'Indicators change colour depending on whether a substance is acid or base.' Possible yellow questions: 'Why is lemon juice sour but soap slippery?' (Otieno and Wanjiru), 'Why do indicators change colour?', 'What does pH 2 mean	Say: 'Take two minutes to write one thing we have learned and one question we still have.' WAIT TIME 10-15 seconds before collecting. Cold-call 4 learners to read and place a note, and ask the class: 'Does this question connect to uses, to safety, or to antacids?' Cluster the questions into themes (why colour changes, what pH means, safety/uses, antacids). Highlight questions that will be answered in Lesson 4 (what makes acids sour; H^+ ions) and later lessons, saying: 'We will	Driving Question Board with Know/Wonder notes: showing both progress and open questions keeps the phenomenon at the centre of the unit and gives learners ownership of the next steps.	Look for: (1) every learner contributes at least one note; (2) at least some questions move from 'what' towards 'why' (e.g. why indicators change colour, why soap is slippery); (3) learners link notes to the driving question on the board; (4) learners connect antacid use to the classification of bases.

 PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	compared with pH 3?', 'If soap is a base, why do antacids also count as bases?', 'Why are patients with hyperacidity given antacids?' Learners link each note to the driving question by drawing an arrow towards 'Does the alkalinity of a substance determine its uses in day to day life?'	come back to this question in a coming lesson.' Keep the board visible for revisiting.		
Model Building Phase  VIDEO: Using a pKa table <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids,Bases and Indicators Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	Learners open their model page: a particle drawing of H⁺ and OH⁻ ions in each product (lemon juice, vinegar, water, baking soda solution, soap solution). Using the classification table and pH values, they make a first draft: lemon juice (pH 2.0) with many H⁺ ions, vinegar (pH 2.8) with fewer H⁺ than lemon, water with equal very few H⁺ and OH⁻, baking soda (pH 8.3) with more OH⁻ than H⁺, soap (pH 10.0) with the most OH⁻. They add a key and a caption in their own words, e.g. 'Lower pH likely means more H⁺'; higher pH likely means more OH⁻' and label which indicator colour matches each beaker. Learners write one 'I wonder' about how the indicator 'knows' the ions are there, to be explored in Lessons 4 and 5.	Say: 'Use your table to decide how many H⁺ and OH⁻ dots to draw. We have not measured ions yet, so mark your drawing as a first idea.' Circulate with prompts: 'Why did you draw more H⁺ in lemon juice than vinegar? What evidence are you using?' and 'How did you show that water is neutral?' WAIT TIME 10-15 seconds after each question. Cold-call 3 learners to explain one beaker to the class. Accept qualitative drawings; do not give exact ion counts or teach the Arrhenius definition in detail, as that is for Lessons 4 and 5. Ask learners to add the label 'Draft 1 - Lesson 3' and keep the model for revision.	Cumulative model building (draft particle model): turning classification data into a visual model helps learners link observable evidence (indicator colours, pH) to particle-level ideas, and starting with a draft encourages revision as new evidence arrives.	Look for: (1) models show more H⁺ for the acids than the bases and more OH⁻ for the bases than the acids; (2) lemon juice is shown as more acidic than vinegar; (3) water is shown as neutral, with equal small amounts of H⁺ and OH⁻; (4) learners label their model as a draft and give a reason from the evidence. Use a 3-2-1 check: 3 products classified, 2 pieces of evidence, 1 remaining question.

D. TEACHER REFLECTION

Reflect: (1) Did learners place water correctly using evidence rather than guessing? (2) Could learners explain why different indicators give different colours but the same classification? (3) Did learners distinguish observation from interpretation when recording colours? (4) Which hypotheses from Lesson 1 were revised, and did any learners resist changing them? (5) Did every learner contribute to the DQB, including quieter learners? (6) Are learners over-interpreting their ion drawings as exact numbers? Plan to revisit any misconceptions at the start of Lesson 4, when the Arrhenius definition of acids is introduced using the H⁺ concentrations of lemon juice and vinegar.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We tested lemon juice, vinegar, water, baking soda solution and soap solution with red cabbage extract and universal indicator.
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	Red cabbage: lemon juice red/pink, vinegar pink-red, water purple, baking soda blue-green, soap green-yellow. Universal indicator: lemon juice red-orange (pH 2), vinegar orange (pH 3), water green (7), baking soda blue (8), soap blue-purple (10).
What did I learn?	Indicators show whether a substance is acidic, neutral or basic. Natural indicators such as red cabbage extract and laboratory indicators such as litmus and universal indicator all give the same sorting, even though the colours differ. Universal indicator also shows how acidic or basic a substance is on the pH scale. Water is neutral (pH 7.0).
How does this explain the phenomenon?	Lemon juice (pH 2.0) and vinegar (pH 2.8) are acids; baking soda (pH 8.3) and soap (pH 10.0) are bases; water (pH 7.0) is neutral. Lemon juice is more acidic than vinegar, and soap is more basic than baking soda. Our hypotheses from Lesson 1 are supported by the indicator evidence and the pH values, and our model shows more H^+ in the acids and more OH^- in the bases. We still need to explain why acids are sour and soap is slippery.

LESSON 4 (40 minutes (one lesson)): Why is Lemon Juice Sour? Acids Release H^+ Ions in Water (Arrhenius Definition)

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners use particle models and a simulation to show that acids release H^+ (aq) ions when dissolved in water, and to use this idea to explain why lemon juice and vinegar are sour, have low pH and turn blue litmus red.
Knowledge	Learners will be able to: (1) state the Arrhenius definition of an acid as a substance that produces H^+ (aq) ions in water; (2) explain that lemon juice (pH 2.0, $H^+ = 1 \times 10^{-2}$ mol/L) has about 6 times more H^+ than vinegar (pH 2.8, H^+ about 1.6×10^{-3} mol/L) and water (pH 7.0, $H^+ = 1 \times 10^{-7}$ mol/L) has far fewer; (3) link a higher H^+ concentration to a lower pH, a stronger sour taste and the litmus and indicator colours seen in Lessons 2-3.
Skills	Developing and using models (particle drawings of H^+ ions); analysing and interpreting data (pH and H^+ concentration); using a simulation; constructing explanations; communicating scientific ideas orally and in writing; collaboration in groups of 4.
Attitudes	Curiosity about why everyday Kenyan foods taste the way they do; respect for evidence and for peers' ideas; safety consciousness (never tasting laboratory chemicals); appreciation of how chemistry explains familiar products from Kisumu market and Kenyan kitchens.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life? (This lesson advances Question 1 by explaining WHY lemon juice and vinegar are acids: they release H^+ (aq) ions in water.)
Purpose in Storyline	Lessons 1-3 established that lemon juice and vinegar sit below pH 7 and were classified as acids using litmus, red cabbage extract and universal indicator. Lesson 4 answers Otieno's question 'Why is lemon sour?' by introducing the particle-level cause: H^+ (aq) ions. Learners add H^+ ions to the Model for lemon juice, vinegar and water. Lesson 5 will build the matching OH^- (aq) idea for bases and alkalis, so bases are NOT explained here.
Safety Notes	No tasting of any laboratory chemical or of any item used in the lesson, even household items (all tasting of lemon juice or vinegar is done only by the teacher as a demonstration, from clean food-grade samples, or referred to from earlier observation). Use only the dilute household samples (lemon juice, vinegar) and wear goggles if learners handle them. Wash hands after handling. Keep the simulation on a shared device or teacher projector if devices are limited; an offline paper simulation card set is provided as an alternative.

B. LESSON OVERVIEW

Mwalimu Achieng revisits the Kisumu kitchen phenomenon: lemon juice (pH 2.0), vinegar (pH 2.8), baking soda (pH 8.3) and bar soap (pH 10.0) look alike, yet lemon juice is sour and soap is slippery. Learners already know from Lessons 1-3 that lemon juice and vinegar are acids with low pH. Today Otieno's question, 'Why is lemon sour?', is explored at the particle level. Learners first predict what lemon juice contains that water does not, then examine H^+ concentration data (lemon juice 1×10^{-2} mol/L, vinegar about 1.6×10^{-3} mol/L, water 1×10^{-7} mol/L) and use a particle simulation or paper-card model to see acids releasing H^+ (aq) ions in water. They explain sourness and low pH using the Arrhenius definition, add the DQB questions, and draw the first H^+ ion version of their Model for each product.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the Kisumu kitchen	Say: 'Remember Otieno asked,	Initial Model Drawing and Public	Look for: (a) drawings that show

 VIDEO: Arrhenius acids and bases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Arrhenius Acids (Flexbook) Source: CK-12  Search ARES for similar readings	display. Mwalimu Achieng holds up a glass of pure water and a glass of lemon juice solution. Learners individually draw, in 3 minutes, what they think the particles look like in each glass, answering: 'What might lemon juice contain that pure water does not, which makes it sour and gives it pH 2.0?' They write a one-sentence prediction and then share with a partner. Typical ideas: 'lemon has acid particles', 'lemon has sugar', 'lemon has more particles'.	why is lemon sour? Both of these look clear. Today we look inside the liquid.' Display the pH graph and point to lemon juice at pH 2.0 and water at pH 7.0. Say: 'Draw what you think is inside each glass. There is no wrong prediction today.' Circulate and note 3 different ideas. Give 3 minutes of silent drawing, then WAIT TIME of 10-15 seconds after asking: 'What did you add to the lemon juice picture that is not in the water picture?' Cold-call 3 learners to share their drawings; do not evaluate, but write each idea on the board under 'Our predictions'. Ask 2 further learners: 'Whose idea is different from yours, and why?'	Prediction Sharing: learners externalise their current ideas about particles in the liquids, so they can be compared with evidence later. This surfaces the misconception that sourness comes from a 'sour particle' or from sugar, and creates a need to know that drives the Observe Phase.	something extra present in lemon juice compared with water; (b) learners who connect pH 2.0 with the picture; (c) misconceptions such as 'lemon juice contains no water' or 'sour comes from sugar'. Tally on a checklist how many of the 3 cold-called learners mention a different particle in lemon juice (target: at least 2 of 3 mention extra particles). Do not correct yet; record ideas for the Explain Phase.
Observe Phase  VIDEO: Arrhenius Acids and Bases - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Arrhenius Acids Practice Source: CK-12  Search ARES for similar readings	Part A (data, 8 min): In groups of 4, learners study a data card: lemon juice pH 2.0, $H^+ = 1 \times 10^{-2}$ mol/L; vinegar pH 2.8, H^+ about 1.6×10^{-3} mol/L; water pH 7.0, $H^+ = 1 \times 10^{-7}$ mol/L. They answer: Which has the most H^+? How many times more H^+ does lemon juice have than vinegar (about 6 times, from a pH difference of 0.8)? How does lower pH relate to H^+ concentration? Part B (simulation or paper model, 10 min): Using an offline particle simulation on a shared device (or paper particle cards if no device is available), learners watch or act out a few acid molecules (citric acid in lemon juice; ethanoic acid (acetic acid) in vinegar) dissolving in water and releasing H^+ (aq) ions, compared with pure water in which almost no H^+ are present. Note: H^+ (aq) is often written H_3O^+; we use H^+ (aq) in this class. Groups record on a sheet: the number of H^+ ions in each	Distribute data cards. Say: 'Look at the numbers. What do you notice about H^+ and pH?' Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 4 groups for one pattern each. Prompt: 'Lemon juice has pH 2.0 and vinegar pH 2.8. The difference is only 0.8, yet lemon juice has about 6 times more H^+. Why do you think a small pH change means a big change in H^+?' (Each pH unit is a 10-fold change.) Introduce the simulation: 'Watch what happens to the acid particles when they meet water. Count the H^+ ions you can see.' Pause the simulation twice and ask: 'What do you see leaving the acid molecule?' Move around checking that each learner draws their own picture. Remind: 'We NEVER taste anything in the laboratory. Sourness is something we explain, not test.' For groups without a device use paper cards: one learner plays water, others play acid molecules that release a	Data Interpretation plus Simulation Observation (Patterns and Models): learners link a numerical pattern (lower pH means higher H^+ concentration) to a visible particle-level process (acid molecules in water releasing H^+ (aq) ions). Combining numbers and visuals builds the cause-and-effect understanding that H^+ ions come from the acid reacting with water.	Check group sheets for: (1) correct ranking of H^+ concentration (lemon juice > vinegar > water); (2) the statement that lemon juice has about 6 times more H^+ than vinegar; (3) particle sketches showing H^+ ions separated from the acid in water. Indicator: at least 3 of 4 groups correctly state that lower pH means more H^+. Note groups that draw H^+ in water too (water has a tiny amount, 1×10^{-7} mol/L, so almost none is the correct picture). Ask any group unsure: 'Which beaker has more H^+, and how do you know from the data?'

	beaker (e.g. many in lemon juice, fewer in vinegar, almost none in water) and sketch each.	card labelled H ⁺ .		
Explain Phase  VIDEO: Arrhenius acids and bases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Acid-Base Solutions Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners work in pairs to write a claim-evidence-reasoning (CER) explanation answering Otieno's question: 'Why is lemon juice sour, and why does it have a lower pH than vinegar?' They use the data card and simulation observations. They then connect the H ⁺ idea to earlier evidence: lemon juice and vinegar turned blue litmus red and the universal indicator showed red-orange and orange (Lessons 2-3). Learners share and refine the class definition: 'An acid is a substance that produces H ⁺ (aq) ions when dissolved in water (Arrhenius definition).' They also note that lemon juice is safe to eat in food because the amount of acid is small and food-grade, but that laboratory acids are dangerous and must never be tasted. Mama mboga uses vinegar, an acid, for pickling kachumbari and onions; learners suggest how a high H ⁺ level might help (preservation, introduced further in Lesson 9).	Say: 'Use a claim, evidence and reasoning to answer Otieno. Your claim must mention H ⁺ ions.' Give 4 minutes of pair writing. Provide the frame: 'Lemon juice is sour because ... Evidence: ... This means ...' Cold-call 3 pairs to read; after each, WAIT TIME of 10-15 seconds, then ask another learner: 'Do you agree? What evidence supports that?' Press for reasoning: 'Why does more H ⁺ give a lower pH?' Then ask: 'Which particle is the same in lemon juice and vinegar that makes both acids?' (H ⁺ (aq).) Clarify: 'Sour taste is how our tongue detects H ⁺ ions, and we explain it with H ⁺ . In the lab we use indicators instead of tasting.' Record the class definition on the board using the Arrhenius wording. Address the misconception that more concentrated always means stronger: say only 'This lesson is about which has more H ⁺ ; we will study strength versus concentration in Lesson 8.' Do not discuss OH ⁻ or bases beyond saying: 'Baking soda and soap are bases; we will explain what particles they have in Lesson 5.'	Claim-Evidence-Reasoning (CER) with Peer Critique: learners must justify the H ⁺ explanation using data and observations, which builds scientific argumentation. Peer challenge ('Do you agree? What evidence?') pushes learners to connect the particle model to macroscopic properties (sour taste, low pH, litmus colour).	Collect pair CER sheets and check against a 3-point rubric: (1) claim mentions H ⁺ (aq) ions released by the acid in water; (2) evidence cites pH or H ⁺ concentration values (lemon juice pH 2.0, 1×10^{-2} mol/L; vinegar pH 2.8, about 1.6×10^{-3} mol/L); (3) reasoning links more H ⁺ to lower pH and stronger sour taste. Target: at least 70% of pairs score 2 or more. Listen for the misconception 'acid means harmful', and respond with the lemon juice and vinegar food examples. Exit-slip option: 'Complete: An acid is a substance that ...'.
Driving Question Board (DQB) Creation  VIDEO: Arrhenius Acids and Bases - Overview Source: CK-12  Search ARES for similar videos	Learners return to the Driving Question Board. In groups, they write on sticky notes (a) one question that today's lesson ANSWERED (e.g. 'Why is lemon sour?') and place it in the 'Answered' column with a one-line answer, and (b) one NEW question raised by today's learning (e.g. 'If soap has no H ⁺ , what particles make it slippery?', 'Why is	Say: 'Look at our DQB. Which questions can we now move to Answered? Which new ones do we have?' Give 3 minutes for writing. WAIT TIME of 10-15 seconds before asking: 'Which question on the board is connected to hyperacidity?' Cold-call 3 groups to place and read one note each. Point out the link: 'Stomach acid is hydrochloric acid,	Driving Question Board Tracking: learners see their growing understanding and the gaps still to fill. Moving answered questions and posting new ones makes the learning trajectory visible and keeps the anchor phenomenon and the hyperacidity question central.	Check that notes (1) correctly identify the answered question about sourness and H ⁺ ; (2) include at least one new, testable question per group; (3) link to the phenomenon (soap, baking soda, antacids). Indicator: at least 4 of 6 groups post a new question that links acids to another product or to hyperacidity. Note questions about bases, strength and antacids as

 HTML: Arrhenius Acids (Flexbook) Source: CK-12  Search ARES for similar readings	vinegar less sour than lemon if both are acids?', 'Why do antacids help if stomach acid has H⁺?'). They link at least one note to the driving question: 'Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications?'	so it also releases H⁺ ions. Keep that idea; it helps us understand antacids in Lesson 7.' Note down new questions that belong to Lesson 5 (bases/OH⁻) and Lesson 8 (strength) and mark them with a coloured dot as 'coming soon'. Do not answer them now.		evidence the storyline is working, and keep them for later lessons.
Model Building Phase  VIDEO: Strong acid solutions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Arrhenius Acids Practice Source: CK-12  Search ARES for similar readings	Learners open their Model page (particle drawings of H⁺ and OH⁻ ions in each product, revised each lesson). They draw three boxes: lemon juice, vinegar and water. In each box they show H⁺ (aq) ions as dots or plus signs (many in lemon juice, fewer in vinegar, almost none in water) and label pH values (2.0, 2.8, 7.0) and H⁺ concentrations (1 x 10⁻², about 1.6 x 10⁻³, 1 x 10⁻⁷ mol/L). They leave the baking soda and soap boxes blank, labelled 'To be added in Lesson 5'. Under the drawings, learners write a caption: 'Lemon juice is sour because the acid releases H⁺ (aq) ions in water.' Finally they complete the Summary Table: what we observed, what we learned and how we explain it.	Say: 'Add today's evidence to your Model. Show the difference between the three liquids using the number of H⁺ ions.' Model one box on the board for lemon juice, then let learners do the rest independently. Walk around and ask: 'How does your drawing show that lemon juice has about 6 times more H⁺ than vinegar?' (For example 6 dots versus 1 dot.) WAIT TIME of 10-15 seconds before cold-calling 3 learners to explain their drawing to the class. Correct any drawings that show H⁺ in water as zero or show lemon juice without water particles. Close by asking: 'What question do you think the soap and baking soda boxes will need to answer?' Collect Models for review. Remind learners to keep the pH graph and Model in their folders for Lesson 5.	Cumulative Model Revision: learners revise a persistent particle model with new evidence from each lesson, which builds a coherent explanation across the unit. Representing quantity by the number of H⁺ ions connects the numerical data (concentration, pH) to the particle-level mechanism and prepares for the OH⁻ ion model in Lesson 5.	Check Model drawings for: (1) H⁺ ions shown in lemon juice, vinegar and water in the correct order of amounts; (2) correct pH and H⁺ values labelled; (3) a caption linking H⁺ release to sourness; (4) honest placeholders for baking soda and soap. Indicator: at least 80% of learners show lemon juice with visibly more H⁺ than vinegar and water with almost none. Use the Summary Table to identify learners needing support, and pair them with a peer for a quick review at the start of Lesson 5.

D. TEACHER REFLECTION

1. Did learners move beyond 'lemon has acid in it' to explain that acids release H⁺ (aq) ions in water? What evidence from the CER sheets supports this? 2. Which learners confused concentration with strength, or thought water has zero H⁺? How will you address this before Lesson 8? 3. Did the simulation or paper cards help learners see ions as separate from the acid molecule? Which groups needed more support? 4. How well did the Kenyan contexts (Kisumu lemon juice, vinegar for kachumbari, stomach hydrochloric acid after nyama choma) keep learners engaged and linked to the driving question? 5. Which DQB questions are ready for Lesson 5 (bases and OH⁻) and Lesson 7 (antacids)? 6. How did you enforce the rule of never tasting laboratory chemicals?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that lemon juice (pH 2.0, $H^+ = 1 \times 10^{-2}$ mol/L) and vinegar (pH 2.8, H^+ about 1.6×10^{-3} mol/L) have much more H^+ than pure water (pH 7.0, $H^+ = 1 \times 10^{-7}$ mol/L), and in the simulation the acid molecules released H^+ (aq) ions when they dissolved in water. Lemon juice has about 6 times more H^+ than vinegar.
What did I learn?	We learned that an acid is a substance that produces H^+ (aq) ions in water (Arrhenius definition), and that a lower pH means a higher H^+ concentration. Each pH unit is a 10-fold change in H^+ concentration.
How does this explain the phenomenon?	We can explain that lemon juice and vinegar are sour, turn blue litmus red and have low pH because the acids (citric acid and ethanoic acid (acetic acid)) release H^+ (aq) ions in water; lemon juice is more sour than vinegar because it has about 6 times more H^+ . Stomach acid is also an acid with H^+ ions, which helps us start to understand why antacids are needed; we still need to learn what particles bases like baking soda and soap release.

LESSON 5 (80 minutes (2 x 40-minute periods)): Arrhenius Definition (Bases and Alkalis): Why Does Soap Feel Slippery?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use evidence from the pH data, indicator results and soap's slippery feel to show that alkalis release OH ⁻ (aq) ions in water, to distinguish a base from an alkali, and to extend their particle model from H ⁺ (aq) in acids to OH ⁻ (aq) in bases.
Knowledge	Define an alkali as a base that dissolves in water producing OH ⁻ (aq) ions (Arrhenius definition). Know that all alkalis are bases but not all bases dissolve in water. Know that soap (pH 10.0) has an OH ⁻ concentration of 1×10^{-4} mol/L (pOH 4), that baking soda (pH 8.3) is only weakly basic, and that the OH ⁻ ions explain why soap feels slippery and why red litmus turns blue.
Skills	Developing and using models (particle drawings of OH ⁻ ions); analysing and interpreting data (pH and OH ⁻ concentration); constructing explanations from evidence; planning and carrying out safe investigations using indicators; communicating scientific ideas orally and in writing.
Attitudes	Curiosity about why everyday products behave as they do; safety consciousness (never taste lab chemicals, wear goggles, wash hands after handling soap solution); respect for others' ideas during discussion; appreciation of chemistry in Kenyan homes, such as Mama Wanjiru's mandazi and cleaning with soap.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life? Lesson focus: What do the basic products (baking soda and soap) have in common at particle level, and why does soap feel slippery?
Purpose in Storyline	Lesson 4 gave the class an H ⁺ (aq) particle model that explains the acids (lemon juice and vinegar). Lessons 2-3 sorted baking soda and soap as bases using litmus, red cabbage and universal indicator. Lesson 5 answers the question these lessons left open: what is in the bases that makes them basic? Learners add OH ⁻ (aq) ions to the model, which prepares for Lesson 6 (reactions of acids) and Lesson 7 (neutralisation, H ⁺ + OH ⁻ giving H ₂ O, and antacids).
Safety Notes	Wear goggles throughout. NEVER taste any lab chemical or the soap and baking soda solutions; tasting is done only through the teacher-led discussion of known observations (soap is bitter and harmful if swallowed). Use dilute soap solution only (1 spoon shavings in 1 cup water). Rinse hands with water if soap solution gets into the eyes or on skin and tell the teacher at once. Do not use sodium hydroxide or lime directly; the teacher shows only the data. Wipe spills immediately, since soap solution makes floors slippery.

B. LESSON OVERVIEW

Mwalimu (Mrs Achieng) in Kisumu returns to the four products from the anchoring phenomenon. Wanjiru asked in Lesson 1 why soap feels slippery, and the class now investigates. Learners predict what soap and baking soda solutions contain, then observe the evidence: red litmus turning blue in both, the pH values (baking soda 8.3, soap 10.0), a feel test of dilute soap solution against vinegar and water, and the OH⁻ concentration of soap (1×10^{-4} mol/L). In the Explain phase, learners use the Arrhenius definition to explain that alkalis release OH⁻ (aq) in water, separate the terms base and alkali, and explain slipperiness. They then add new questions to the Driving Question Board and revise the particle model to show OH⁻ (aq) in baking soda and soap alongside the H⁺ (aq) of lemon juice and vinegar. This brings the class closer to answering: Does the alkalinity of a substance determine its uses, and why do hyperacidity patients get antacids?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Arrhenius acids and bases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: pH Scale Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners recall the phenomenon graph: lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0. Wanjiru's question is revisited: why does soap feel slippery? Individually, learners draw a quick particle prediction in their notebooks of what is in a cup of soap solution compared with a cup of lemon juice (using the H ⁺ model from Lesson 4 as a start). They write one sentence: 'I think soap solution feels slippery because...'. Then they share with a partner and add a confidence rating from 1 to 3.	Display the four products and the pH graph. Say: 'In Lesson 4 we showed that lemon juice and vinegar release H ⁺ ions in water. Today we look at the other side of 7. Wanjiru asked why soap feels slippery. What might be in soap solution that is not in lemon juice?' Give 2 minutes of silent drawing. WAIT TIME: 10-15 seconds after the question before taking any answer. Then cold-call 3 learners (not volunteers) to describe their drawing; ask a fourth learner to restate one drawing in their own words. Say: 'I am not looking for the right answer yet; I want your best thinking.' Record the predictions on chart paper without judging, and keep the chart for the later phases.	Predict-Share-Revisit (Developing and using models): committing to a prediction in drawing form makes learners' existing ideas visible. Later they compare it with the evidence, which creates a reason for the revision and builds ownership of the final model.	Listen and look for: (a) learners who draw only H ⁺ in soap, or no particles at all; (b) learners who suggest a different particle in bases, since this shows readiness for OH ⁻ ; (c) learners who link the pH above 7 to something being 'opposite' to acids. Tally on the clipboard how many predictions mention a new particle. Note confidence ratings and revisit them at the end of the lesson.
Observe Phase  VIDEO: Arrhenius Acids and Bases - Overview Source: CK-12  Search ARES for similar videos  HTML: Arrhenius Bases (Flexbook) Source: CK-12  Search ARES for similar readings	Stations in groups of 4 (rotating 5 minutes each), wearing goggles. Station 1 - Litmus: red and blue litmus in baking soda solution, soap solution, vinegar and water; record colours (red litmus turns blue in baking soda and soap; blue litmus stays blue; vinegar turns blue litmus red). Station 2 - Feel test: the teacher-approved test of rubbing a drop of dilute soap solution between finger and thumb, compared with a drop of water and a drop of vinegar; learners describe the feel in words (slippery vs not). Learners NEVER taste anything. Station 3 - Data card: the pH values (baking soda 8.3, soap 10.0, water 7.0), and the OH ⁻ concentration of soap, 1 x 10 ⁻⁴ mol/L (pOH 4), against the water value of 1 x 10 ⁻⁷ mol/L for both H ⁺ and OH ⁻ . Learners calculate that soap has 1000 times	Before the stations, say: 'Remember our rule: goggles on, no tasting any chemical. If soap solution gets near your eyes, tell me at once.' Circulate and ask each group: 'What do baking soda and soap have in common in your results?' and 'What is different between soap and baking soda?' At Station 3 ask: 'Soap is pH 10.0 and baking soda 8.3. Which has more OH ⁻ and how do you know?' WAIT TIME: 10-15 seconds after each question. Cold-call 2 learners from different groups at the end of each rotation to give one observation. Say: 'Point to the data that supports what you are saying.' Support learners who struggle with powers of ten by writing 10 ⁻⁴ as 0.0001 and 10 ⁻⁷ as 0.0000001 on the board.	Evidence Table and Data Interpretation (Analysing and interpreting data): collecting different kinds of evidence (colour change, touch, pH, ion concentration) in one table lets learners find the pattern that all the bases share, so the explanation grows from data rather than from being told.	Check that each group's table shows: red litmus turning blue in baking soda and soap, the correct pH values (8.3 and 10.0), and the 1000-times calculation. Observable indicators: learners compare baking soda and soap using pH numbers and not only words; learners point at the data when explaining. Note groups that mix up pH and OH ⁻ concentration for follow-up in the Explain phase.

	more OH ⁻ than pure water (10 ⁻⁴ / 10 ⁻⁷). Station 4 - Reading card: soap is made from fats or oil and sodium hydroxide, and the OH ⁻ ions in soap solution react with the oils on skin, which creates the slippery feel. Learners record observations in a table: substance, litmus result, feel, pH, comment.			
Explain Phase  VIDEO: Arrhenius acids and bases Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Arrhenius Bases Practice Source: CK-12  Search ARES for similar readings	Whole-class discussion followed by writing. Learners answer Wanjiru's question using the Claim-Evidence-Reasoning structure: Claim (what soap solution contains that makes it slippery and basic), Evidence (litmus, pH 10.0, OH ⁻ of 1 x 10 ⁻⁴ mol/L, slippery feel, no slipperiness in vinegar), and Reasoning (the Arrhenius definition: bases that dissolve in water, called alkalis, produce OH ⁻ (aq) ions; the OH ⁻ ions react with oils on the skin to give a slippery feel). Learners then explain the difference between a base and an alkali using a sorting activity with cards: sodium hydroxide (alkali), calcium hydroxide (alkali), ammonia solution (alkali), magnesium hydroxide (base, hardly dissolves), and copper(II) oxide (base, insoluble). They conclude: all alkalis are bases but not all bases are alkalis. They explain baking soda (pH 8.3) as a weak base: it gives only a few OH ⁻ ions and so is only slightly above neutral and safe in small amounts in mandazi. The mention of H ₃ O ⁺ is only a note: the H ⁺ (aq) from Lesson 4 is really H ₃ O ⁺ in water, and the class continues with H ⁺ (aq).	Say: 'Wanjiru asked why soap feels slippery. Using the evidence on your table, write a claim.' Give 3 minutes of writing. Cold-call 3 learners to read their claim; after each, ask: 'What evidence supports that?' and 'Who can add reasoning using the word OH ⁻ ?' WAIT TIME: 10-15 seconds. Write the definition on the board: 'Base: substance that neutralises acids. Alkali: a base that dissolves in water producing OH ⁻ (aq) ions.' Then use the card sort: 'Where does magnesium hydroxide go? It is in antacids; we will see why in Lesson 7.' Do NOT teach neutralisation now; just say: 'Bases react with acids, and we will test this in Lesson 7.' Ask: 'Soap is more basic than baking soda. Using OH ⁻ ions, how would you explain that?' Correct misconceptions: pH above 7 does not mean strong; strength is Lesson 8. Do not explain strong vs weak in detail, only that baking soda gives fewer OH ⁻ than soap.	Claim-Evidence-Reasoning with Card Sort (Constructing explanations): learners connect observations to the particle-level idea (OH ⁻ ions) and use the sort to see the boundaries of the new vocabulary, so 'base' and 'alkali' become meaningful categories rather than memorised words.	Look for: claims that name OH ⁻ (aq) ions; reasoning that links OH ⁻ to alkalis dissolving in water; correct sorting of at least 4 of 5 cards; the sentence 'all alkalis are bases but not all bases are alkalis'. Common errors to note: saying soap contains 'no H ⁺ ', saying 'bases are the opposite of acids' without the ion idea, or confusing alkali with any base. Use a thumbs-up check: 'Show me 1 to 3 fingers on how confident you are in the difference between a base and an alkali.'
Driving Question	Learners go to the Driving Question Board. They add sticky	Say: 'Look at our DQB. Which questions can we now answer with	Driving Question Board (Asking questions): returning to learners'	Check that at least half of the new questions are about interaction

Board (DQB) Creation  VIDEO: Arrhenius Acids and Bases - Overview Source: CK-12  Search ARES for similar videos  HTML: Dropping Base, Arrhenius Base (PLIX) Source: CK-12  Search ARES for similar readings	notes with answers to earlier questions (e.g., Wanjiru's question about slippery soap, marked 'answered: OH- ions react with oils on skin') and write new questions they now have. Example questions: What happens when H⁺ from lemon juice meets OH⁻ from soap? Why are antacids bases? Why is baking soda safe in mandazi but soap harmful? Why does vinegar react with baking soda and fizz? How do strong and weak bases differ? Learners place their questions under categories: Reactions, Strength, Uses, Safety.	evidence? Which new questions did today's lesson raise?' Give 3 minutes for writing. WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 3 learners to read a question aloud and place it. Ask one learner to explain why a question moves to 'answered'. Link back to the driving question: 'Does alkalinity determine how soap and baking soda are used? What do we still need to find out about antacids?' Point out which lesson will address each question: reactions in Lesson 6, neutralisation and antacids in Lesson 7, strength in Lesson 8.	questions shows them how their understanding is progressing and keeps the investigation anchored in their curiosity about the phenomenon.	between H⁺ and OH⁻ (neutralisation), strength, or the safety of products. Observable indicator: learners referring to evidence from today when marking Wanjiru's question as answered. Note learners whose questions are still only about recalling facts and prompt them: 'What would you test to find out?'
Model Building Phase  VIDEO: Strong base solutions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: pH Scale Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners revise their particle models from Lesson 4. On a page divided into four boxes (lemon juice, vinegar, baking soda solution, soap solution), they draw H⁺ (aq) as the particle in lemon juice (1 x 10⁻² mol/L) and vinegar (about 1.6 x 10⁻³ mol/L) from Lesson 4, and now add OH⁻ (aq) to baking soda (fewer ions, pH 8.3) and soap (more ions, 1 x 10⁻⁴ mol/L, pH 10.0). They use a key (+ for H⁺, - for OH⁻) and show relative numbers of particles, with a short caption linking each product to its use: vinegar for pickling, soap for cleaning, baking soda for raising mandazi, lemon juice for flavour. They add a note: 'Soap feels slippery because OH⁻ ions react with oils on skin.' Learners compare with their Predict drawing, using a different colour pen to show what has changed, and write one sentence: 'My model changed because...'. Finally they draw a question mark between a H⁺ box and an OH⁻ box: 'What happens when they meet?' to prepare for Lesson 7.	Say: 'Take out your Predict drawing. Now use today's evidence to build the next version of the model. Every particle you draw must be backed by evidence.' Circulate and ask: 'Why did you draw more OH⁻ in soap than in baking soda?' and 'What evidence tells you that?' WAIT TIME: 10-15 seconds. Cold-call 3 learners to explain one box of their model; invite 1 learner to share a change from their Predict drawing. Model the comparison: 'In Lesson 4 our model only had H⁺. What does the model need now?' Collect the models and keep them for Lesson 6. Close by linking back: 'We have two kinds of particles that explain the products. Next, in Lesson 6, we find out how the H⁺ in acids behaves with metals and carbonates, and in Lesson 7 we see what happens when H⁺ meets OH⁻, which is what antacids do for a patient with hyperacidity.'	Cumulative Model Revision (Developing and using models): revising a model against the earlier prediction makes the learning visible, so learners can see how evidence changed their thinking and use the model to predict future behaviour in Lessons 6 and 7.	Check models for: H⁺ in lemon juice and vinegar with more in lemon juice; OH⁻ in baking soda and soap with more in soap; no H⁺ or OH⁻ confusion; a caption linking OH⁻ to slippery feel; a visible change from the Predict drawing. Exit ticket: 'Write one sentence defining an alkali and name one product from our lesson that contains one.' Success criterion: the answer mentions a base that dissolves in water producing OH⁻ (aq) and names soap (or baking soda solution). Reteach in the next lesson's warm-up for learners who write 'a substance with pH above 7' only.

D. TEACHER REFLECTION

Which learners could explain the slippery feel using OH⁻ ions and which only repeated 'soap is basic'? Did the card sort make the base versus alkali distinction clear, or do some learners still use the words interchangeably? Were the powers of ten manageable for learners (10⁻⁴ versus 10⁻⁷), or do I need a number line next time? Did every learner's revised model show more OH⁻ in soap than in baking soda, and did they justify it with the pH data? Did I keep to the rules of the lesson map by leaving neutralisation for Lesson 7 and strength for Lesson 8? How well did safety routines (goggles, no tasting) hold during the feel test?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Red litmus turned blue in baking soda solution and soap solution, but stayed red in lemon juice and vinegar. Soap solution (pH 10.0) felt slippery, baking soda solution (pH 8.3) was only weakly basic, and soap has an OH ⁻ concentration of 1 x 10 ⁻⁴ mol/L, 1000 times more than pure water (1 x 10 ⁻⁷ mol/L).
What did I learn?	Bases that dissolve in water are called alkalis and they produce OH ⁻ (aq) ions (Arrhenius definition). All alkalis are bases but not all bases dissolve in water. Soap has more OH ⁻ ions than baking soda, so it has a higher pH.
How does this explain the phenomenon?	Soap feels slippery because it is an alkali that releases OH ⁻ (aq) ions in water, and these ions react with the oils on the skin. Baking soda gives fewer OH ⁻ ions, so it is only weakly basic and safe in small amounts in mandazi. Our particle model now shows H ⁺ (aq) in lemon juice and vinegar and OH ⁻ (aq) in baking soda and soap.

LESSON 6 (80 minutes (2 x 40-minute periods)): Chemical Properties I: How Acids React with Metals and Carbonates

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how acids react with a reactive metal and with carbonates, and use gas tests to identify the gases produced. This helps explain why vinegar fizzes with baking soda in the Kisumu kitchen and how the H^+ (aq) ions in acids behave in chemical reactions.
Knowledge	Acids react with reactive metals such as magnesium to give a salt and hydrogen gas. Acids react with carbonates and hydrogencarbonates to give a salt, water and carbon dioxide gas. Hydrogen burns with a squeaky pop. Carbon dioxide turns limewater milky. Both reactions depend on the H^+ (aq) ions in the acid.
Skills	Planning and carrying out an investigation (NGSS SEP 3). Analysing and interpreting data from observations (SEP 4). Developing and using models: extending the particle drawing of H^+ ions to show what happens in reactions (SEP 2). Constructing explanations from evidence (SEP 6). Writing word equations.
Attitudes	Curiosity about why the kitchen products fizz. Care and safety when handling acids and flammable gas. Respect for evidence when it contradicts a prediction. Co-operation in group work.
Key Inquiry Question	How do we classify substances as acids or bases? Why are acids and bases important in our day-to-day life?
Purpose in Storyline	Lessons 4-5 taught that acids release H^+ (aq) ions in water and that bases and alkalis release OH^- (aq) ions. Lesson 6 shows what these H^+ ions DO chemically: they react with metals and carbonates, giving gases that can be tested. This gives learners a chemical test (not just a colour change or a taste) to classify lemon juice and vinegar as acids, and it explains why vinegar and baking soda fizz. It prepares Lesson 7, where H^+ ions react with OH^- ions in neutralisation and antacids.
Safety Notes	Wear goggles throughout. Never taste any lab chemical. Use only dilute (0.1 M) hydrochloric acid and small 2 cm strips of magnesium. Hydrogen is flammable: collect it in a small test tube only, and bring the lighted splint to the mouth of the tube, away from the face and with other flammable items kept away. Teacher demonstrates or closely supervises the squeaky pop test. Limewater is an irritant: avoid contact with eyes and wash hands after use. Wipe up spills at once and rinse skin with plenty of water. Dispose of waste in the labelled container.

B. LESSON OVERVIEW

Mrs Achieng returns to the four products on display in the Kisumu classroom and the pH graph (lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0). She mixes vinegar with baking soda (the mandazi ingredient) and the mixture fizzes. Learners ask what the gas is and why it happens. Learners predict, then investigate in groups: (1) vinegar + baking soda with a limewater test, (2) magnesium ribbon + dilute hydrochloric acid with a lighted splint test. They connect the fizzing to the H^+ (aq) ions from Lesson 4 and to baking soda as a carbonate-type base from Lesson 5, write word equations, and extend the particle model. They add new questions to the Driving Question Board (DQB) and revise their model of H^+ and OH^- ions in each product. This advances the driving question by showing that the acidic nature of a substance, due to H^+ ions, determines how it behaves chemically and therefore how it is used.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look at the display of	Hold up the cup of vinegar and the	Predict-Observe-Explain (POE)	Check the notebooks for: (1) a

 VIDEO: 03 Acid And Metal Carbonate Reactions Source: Seavuria  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: acids bases metal carbonate	lemon juice, vinegar, baking soda and soap with the pH graph. Mrs Achieng adds a spoon of baking soda to a cup of vinegar and the mixture fizzes strongly. Mama Wanjiru uses baking soda in mandazi, so learners link the fizz to cooking. Individually, in their notebooks, learners write: (a) what do you think the gas is? (b) will lemon juice also fizz with baking soda? (c) what do you predict will happen if a strip of magnesium ribbon is placed in dilute hydrochloric acid, the acid in our stomachs? They draw a quick sketch of the H⁺ ions they think are involved. Then they share predictions in pairs.	spoon of baking soda. Say: "Before I mix these, I want you to predict. What will you see, and what could the gas be?" Mix and let learners observe for 10 seconds. Say: "Otieno asked earlier why lemon is sour. Today we ask: what do the H⁺ ions in acids actually DO?" Give 2 minutes of silent writing. WAIT TIME: after asking "What could the gas be?" wait 10-15 seconds before taking any answers. Cold-call 4 learners, including one from the back row, to give their prediction and reason. Record all predictions on the board without judging them, including ideas such as air, oxygen, steam, carbon dioxide or hydrogen. Ask: "Which prediction would you like to test first, and how could we test it?"	with a Quick Write: committing to a written prediction makes learners aware of their prior ideas, so that evidence which confirms or contradicts them is meaningful. Recording the class predictions makes the ideas public and gives a reason to investigate.	prediction with a reason, not a blank or a bare guess; (2) any link to the H⁺ ions from Lesson 4 in their sketch; (3) whether learners suggest a way to test the gas. Tally on the board how many predict carbon dioxide, oxygen, hydrogen or air, to identify misconceptions (for example that all gases from reactions are oxygen).
Observe Phase  VIDEO: 01 Acid And Base Reactions With Metals Source: Seavuria  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: acids bases metal carbonate	Groups of 4 carry out two investigations with a data table. Investigation A (carbonate): add 1 spoon of baking soda to 10 mL of vinegar in a test tube. Use a delivery tube to pass the gas into limewater (or hold a tube of limewater at the mouth). Record the fizzing and the limewater colour change (it turns milky). They then repeat with lemon juice to see if it also fizzes, and test water + baking soda as a control (no fizz). Investigation B (metal): place a 2 cm strip of magnesium ribbon in 5 mL of 0.1 M dilute hydrochloric acid in a test tube. Record the bubbles and warmth. Cover the tube with a thumb or stopper to collect the gas, then bring a lighted splint to the mouth of the tube (teacher-supervised) and listen for the squeaky pop. Each group also tries magnesium in water as a control (no bubbles).	Circulate and say: "Describe only what you see and hear. Do not tell me the gas yet." Check that every learner wears goggles: "Goggles on before the acid comes out." For the splint test, stand beside each group: "Hold the tube with the mouth down and bring the splint to the mouth, away from your face." Ask probing questions: "What is different between the tube with vinegar and the tube with water?" and "Why do we need a control?" WAIT TIME: after asking "What does the milky limewater tell us?" wait 10-15 seconds before accepting answers. Cold-call 3 groups to report their limewater result and 3 groups to report the splint result. If results differ, ask: "What could have caused this difference: amount, age of the magnesium, or the acid?" Remind: "Never taste anything in the lab, even vinegar or baking soda used	Structured observation with controlled comparison: using water as a control for each test lets learners see that the fizzing needs an acid, building the understanding that the acid is the cause. Separating observation from interpretation (describing before naming) prevents jumping to conclusions and strengthens the evidence base.	Observable indicators: (1) learners record 'fizzing/bubbles' and 'limewater turned milky' accurately; (2) learners record 'squeaky pop' for the magnesium + HCl tube; (3) learners note that controls (water) did not produce the gas; (4) safe handling: goggles on, splint held away from the face. Use a checklist while circulating. Learners who record 'explosion' or 'smoke' instead of a small pop need follow-up questions.

	Learners record observations as: substances used, what was seen, what was heard, test result.	here."		
Explain Phase  VIDEO: 03 Acid And Metal Carbonate Reactions Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: acids bases metal carbonate	Each group uses its data to answer: What gas was produced in each reaction, and how do we know? Learners write the claim-evidence-reasoning for each reaction. They complete word equations: vinegar (ethanoic acid, also called acetic acid) + sodium hydrogencarbonate gives a salt + water + carbon dioxide; magnesium + hydrochloric acid gives magnesium chloride + hydrogen. They then write the general patterns: acid + metal gives salt + hydrogen; acid + carbonate gives salt + water + carbon dioxide. Learners link back to Lesson 4 (acids release H ⁺ (aq) ions) and Lesson 5 (baking soda as a carbonate-type base) to explain why the H ⁺ ions from the acid are needed for both reactions. They discuss why lemon juice and vinegar fizz with baking soda, why this does not happen with soap solution, and why mandazi dough rises when baking soda meets an acid in the dough. They compare with the pH graph: lemon juice (pH 2.0) has about 6 times more H ⁺ than vinegar (pH 2.8), so they predict which will fizz faster and check against their data.	Say: "Use your evidence. Tell me: I claim..., my evidence is..., this means..." Model one claim-evidence-reasoning on the board for the limewater test. Ask: "Where did the H ⁺ ions we drew in Lesson 4 go in these reactions?" WAIT TIME: wait 10-15 seconds, then ask pairs to talk for 1 minute before answering. Cold-call 4 learners to explain one test and 2 learners to state the general pattern in their own words. Probe: "Magnesium is a metal and baking soda is a carbonate. What is the same about both reactions?" (the acid supplies H ⁺). Ask Wanjiru's type of question: "Why does soap not give fizz with magnesium?" and guide learners to see it is not an acid. Correct misconceptions: the gas from a metal and acid is hydrogen, not oxygen; the fizzing in the kitchen is carbon dioxide, not air. Introduce, as a note only, that H ⁺ (aq) is often written H ₃ O ⁺ . Do not teach neutralisation here; say: "Next lesson we ask what happens when an acid meets a hydroxide base such as the one in antacids."	Claim-Evidence-Reasoning (CER) with Pair Talk: requiring each claim to be supported by a specific observation (milky limewater, squeaky pop) and a scientific reason (H ⁺ ions react) builds evidence-based explanation. Pair talk lets learners rehearse ideas before sharing publicly, so more learners take part.	Look for: (1) each claim naming the correct gas (carbon dioxide for the carbonate, hydrogen for the metal); (2) evidence tied to the correct test (limewater milky; squeaky pop); (3) correct word equations; (4) reasoning that links the reaction to H ⁺ (aq) ions in the acid. Ask 3 learners to complete an exit sentence: 'Acids react with metals to give ____ and with carbonates to give ____.' Note learners who confuse the two gas tests.
Driving Question Board (DQB) Creation  VIDEO: 01 Acid And Base Reactions With Metals Source: Seavuria  Search ARES for similar videos	Learners return to the Driving Question Board with the question: Does the alkalinity of a substance determine its uses in day-to-day life? Why are patients with hyperacidity issues given antacid medications? Using sticky notes, each group writes (a) one question that today's evidence helps to answer, and (b) one new question.	Say: "Let us look at our Driving Question Board. Which questions can we now answer with today's evidence?" Move answered notes across with the class, for example 'Why does vinegar fizz with baking soda?' WAIT TIME: after asking "What new question does today's evidence raise?" wait 10-15 seconds. Cold-call 3 groups to	Driving Question Board tracking: sorting questions into answered, partly answered and new shows learners their own growth in understanding and keeps the investigation learner-driven, linking each activity back to the phenomenon and the driving question.	Check that: (1) each group writes at least one question and one new question; (2) questions move beyond recall to 'why' or 'how' questions; (3) at least some learners link acid reactions to antacids, safety or uses; (4) learners can say which DQB questions were answered by today's evidence and give the

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: acids bases metal carbonate	Examples: Why does the fizzing in the vinegar and baking soda stop? Can stomach acid react with metals? What happens when an acid meets a base like magnesium hydroxide? Is it safe that antacids containing calcium carbonate fizz in the stomach? Why do car batteries and cleaning products have to be handled carefully? Learners place the notes on the DQB under categories such as 'Answered', 'Partly answered' and 'New'.	read and place their new question. Point to a question about antacids or hyperacidity: "Today we saw that acids react with carbonates. Calcium carbonate is in some antacid tablets. What do you think happens in the stomach? We will explore this in Lesson 7." Highlight the questions that connect acid reactions to safety and uses (for example why sulphuric acid in car batteries or strong acids must be handled carefully).		supporting observation.
Model Building Phase  VIDEO: Electrolyte test (pure water vs. vinegar) Related topic: not an exact match for this lesson. Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: acids bases metal carbonate	Learners revise the particle model they began in earlier lessons (H^+ and OH^- ions in each product). Individually, then in groups, they add to the drawings: (1) in vinegar and lemon juice, H^+ (aq) ions shown reacting with the carbonate in baking soda and leading to carbon dioxide gas bubbles and water; (2) in the hydrochloric acid and magnesium tube, H^+ ions combining to form hydrogen gas bubbles while magnesium goes into solution as a salt; (3) in soap solution, many OH^- ions and no reaction that gives fizz with magnesium. They add a key showing H^+, OH^-, CO_2, H_2 and a caption for each product. They add a short written explanation: 'Why does vinegar fizz with baking soda?' and 'What does this tell us about vinegar?' Groups do a gallery walk, leaving one comment on a neighbouring model.	Say: "Add what you learned today to your model. Show me the H^+ ions and where the gas comes from." Provide the prompt: "Your model must explain the fizzing in the Kisumu kitchen." Circulate and ask: "Where are the H^+ ions before and after the reaction?" WAIT TIME: give 10-15 seconds of think time before pairs discuss and revise. Cold-call 3 groups to explain their model to the class. For the gallery walk, set the rule: "One star (what is clear) and one question (what is unclear)." Close by asking: "How does today's model help us answer the driving question about uses of acids and bases?" and guide learners to see that the chemical behaviour of acids (reacting with metals and carbonates) is part of why acids are used and handled in particular ways. Do not introduce neutralisation or strong versus weak ideas here; those come in Lessons 7 and 8.	Cumulative model revision with Gallery Walk peer feedback: revising the same particle model each lesson shows how new evidence refines understanding, and peer star/question comments push learners to justify their drawings.	Look for models that: (1) show H^+ ions in vinegar, lemon juice and hydrochloric acid; (2) show gas bubbles labelled carbon dioxide or hydrogen; (3) show that soap and baking soda have OH^- (or carbonate-type) behaviour consistent with Lessons 4-5; (4) include a written explanation using the words H^+ (aq), carbonate, metal and gas. Use a 3-level rubric: beginning (gas drawn, no H^+ link), developing (H^+ linked to one reaction), secure (H^+ linked to both reactions with correct gases).

D. TEACHER REFLECTION

After the lesson, consider: Did learners distinguish the two gas tests (milky limewater for carbon dioxide, squeaky pop for hydrogen)? Which predictions changed after

the evidence, and why? Were safety practices followed during the splint test, and does any group need extra support with goggles or handling? Did learners connect the fizzing to H^+ (aq) ions from Lesson 4 and to baking soda from Lesson 5 rather than treating it as a separate fact? Which DQB questions about antacids and hyperacidity are ready for Lesson 7? Note learners who still confuse acids reacting with metals and with carbonates, and plan a short retrieval task at the start of the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Vinegar mixed with baking soda fizzed, and the gas turned limewater milky. Lemon juice also fizzed with baking soda, but water with baking soda did not. Magnesium ribbon in dilute hydrochloric acid gave bubbles, and the gas burned with a squeaky pop. Magnesium in water showed no bubbles.
What did I learn?	Acids react with reactive metals such as magnesium to give a salt and hydrogen gas, and with carbonates such as baking soda to give a salt, water and carbon dioxide. Hydrogen is tested with a lighted splint (squeaky pop) and carbon dioxide with limewater (turns milky). These reactions happen because of the H^+ (aq) ions in acids.
How does this explain the phenomenon?	The fizzing in the Kisumu kitchen happens because vinegar and lemon juice are acids that release H^+ (aq) ions, which react with the carbonate-type baking soda to give carbon dioxide. Soap does not do this because it is a base. This shows that the acidic or basic nature of a substance decides how it behaves and is used, which helps us answer the driving question and prepares us to explain how antacids deal with hyperacidity.

LESSON 7 (80 minutes (2 x 40-minute periods)): Chemical Properties II: Neutralisation and Antacids

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate what happens when an acid meets a base, and use the evidence to explain how antacids relieve hyperacidity and why alkalinity determines the use of a substance.
Knowledge	Neutralisation: acid + base gives salt + water. In ion terms, H^+ (aq) + OH^- (aq) gives H_2O (l). Stomach acid is hydrochloric acid with pH about 1.5 to 2. An antacid containing a base such as magnesium hydroxide or calcium carbonate neutralises some of this acid and raises stomach pH to about 3.5 to 5. Magnesium hydroxide + hydrochloric acid gives magnesium chloride + water.
Skills	Planning and carrying out an investigation (NGSS SEP: Planning and Carrying Out Investigations); analysing and interpreting data (SEP); developing and using models (SEP); constructing explanations (SEP). Learners measure colour or pH change, record observations, and write word and ion equations.
Attitudes	Curiosity about how medicines work, care and responsibility when handling chemicals and medicines, respect for evidence, and cooperation in group work.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life?
Purpose in Storyline	Lessons 4-6 gave us the H^+ ion model for acids, the OH^- ion model for alkalis (Lesson 5) and the reactions of acids with metals and carbonates (Lesson 6). Lesson 7 shows what happens when H^+ meets OH^- . This is the key evidence needed to answer the second part of the driving question: why patients with hyperacidity are given antacids. Lesson 8 will then ask how strength affects the pH change.
Safety Notes	Wear goggles. Never taste any laboratory chemical or the tablets used in the investigation; the antacid tablet is used as a lab material only. Use only very dilute (0.1 M) hydrochloric acid and limewater/magnesium hydroxide suspension, handled by the teacher where possible. Wash spills with plenty of water. Wash hands after the practical. In real life, medicines are taken only as directed by a nurse or pharmacist.

B. LESSON OVERVIEW

Mwalimu Achieng at a school in Kisumu reminds learners of the four products from the pH graph: lemon juice (pH 2.0), vinegar (pH 2.8), baking soda solution (pH 8.3) and soap solution (pH 10.0), with water at 7.0. Learners now meet a new puzzle: after eating spicy nyama choma in Nairobi, a patient has hyperacidity and the pharmacist gives a magnesium hydroxide antacid. Learners predict what happens to the pH when a base is added to an acid. They then observe a neutralisation investigation using dilute hydrochloric acid (as a model of stomach acid), universal indicator and magnesium hydroxide / baking soda, and watch the colour move from red-orange towards green as the pH rises. They explain the change with H^+ and OH^- particles forming water, write the word equation for the reaction, and connect the stomach pH change (about 1.5 to 2 rising to about 3.5 to 5) to relief from hyperacidity. They then add to the Driving Question Board and revise their particle models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the pH graph of lemon juice (2.0), vinegar (2.8),	Display the pH graph and say: 'Remember the four products from	Predict-Share-Revise (individual prediction, then partner talk).	Observe whether learners (a) link the antacid to a base from Lesson

 VIDEO: Neutralization Reaction - Overview Source: CK-12  Search ARES for similar videos  PDF: Acids Bases and salts Source: ARES Local Curriculum - Digital Literacy Content  Search ARES for similar readings	water (7.0), baking soda (8.3) and soap (10.0). Mwalimu Achieng tells the story of a patient in Nairobi who ate spicy nyama choma and now has a burning feeling in the stomach (hyperacidity). The pharmacist gives a magnesium hydroxide antacid. Learners individually draw a quick prediction in their notebooks: what will happen to the H⁺ ions in stomach acid (hydrochloric acid, pH about 1.5 to 2) when the antacid is added, and what will happen to the pH? They then share with a partner and then the group.	our kitchen in Kisumu. Today we meet a medicine that is also a base.' Ask: 'A patient has too much acid in the stomach. The pharmacist gives magnesium hydroxide, a base. Draw and write what you think will happen to the acid and to the pH.' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before asking for answers. Cold-call 4 learners (including Otieno and Wanjiru's groups) to share predictions and ask a different learner to restate each idea in their own words. Record all predictions on the board without judging them as right or wrong. Ask: 'Which earlier idea from Lessons 4 and 5 (H⁺ in acids, OH⁻ in alkalis) helped you make that prediction?'	Writing a prediction before seeing evidence makes learners commit to their current ideas about H⁺ and OH⁻ ions, so that later evidence can challenge or confirm those ideas and the conceptual change is visible.	5, (b) predict the pH rises, and (c) use H⁺ or OH⁻ particles in their drawings. Look for misconceptions such as 'the antacid soaks up the acid like a sponge' or 'the base just mixes with the acid and nothing changes'. Note the number of learners (aim for at least 4 out of 10 sampled) who mention ions in their drawing and record the common ideas for revisiting after the Observe Phase.
Observe Phase  VIDEO: Neutralization Reaction - Overview Source: CK-12  Search ARES for similar videos  HTML: Acid-Base Neutralization (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4, learners carry out a neutralisation investigation. Model stomach acid: 20 mL of 0.1 M hydrochloric acid in a beaker with 5 drops of universal indicator (red, pH about 1). Step 1: add magnesium hydroxide suspension (or a crushed antacid tablet) a little at a time with stirring. Learners record the colour after each addition: red, orange, yellow and finally green or blue-green as the pH rises. Step 2 (comparison): repeat with baking soda solution (pH 8.3) added to the acid, noting the fizzing as CO₂ forms (Lesson 6) and the colour change. Step 3 (control): add the same volume of water to the acid and note that the colour changes very little. Learners record results in a table of added amount, colour and estimated pH, and then plot a simple graph of pH against amount of base added. Teacher also shows the data: stomach pH about 1.5 to 2 before	Before starting say: 'Goggles on. We do not taste anything in the lab.' Demonstrate how to add base slowly and stir. Circulate and ask each group: 'What colour is the indicator now, and what pH does the chart say?' and 'What is different between adding water and adding the base?' Give WAIT TIME of 10-15 seconds after each question. Stop the class after Step 3 and cold-call 3 groups to report the colour sequence. Ask Wanjiru's group to explain what the fizzing with baking soda tells us (carbon dioxide from a carbonate-type base, Lesson 6). Show the stomach pH data on the board: 'Before antacid pH 1.5-2; after antacid pH 3.5-5. Compare this with our beaker.' Ask: 'Did the pH go all the way to 14? Why might that matter for a real stomach?'	Data table and graph analysis (pattern finding). Recording colour and pH after each addition lets learners see a pattern: the more base added, the higher the pH. Comparing with the water control shows that the change is caused by the base and not just dilution.	Check the data tables for accurate colour-to-pH matching using the universal indicator chart, a control comparison and a graph with labelled axes. Indicators of success: learners report that the pH rose from about 1 towards 7 as the base was added and that water alone changed it very little; learners identify the fizzing as CO₂ with baking soda. Ask 3 learners to say what stayed the same and what changed. Note any group that misreads the colour chart and re-teach with the chart.

	antacid and about 3.5 to 5 after.			
Explain Phase  VIDEO: Titration of a strong acid with a strong base <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acid-Base Neutralization Practice Source: CK-12  Search ARES for similar readings	Learners use their evidence to explain neutralisation. In groups they draw particle diagrams before and after: hydrochloric acid with many H^+ (aq) ions, magnesium hydroxide providing OH^- (aq) ions, and then H^+ (aq) + OH^- (aq) joining to form H_2O. They write the word equation: hydrochloric acid + magnesium hydroxide gives magnesium chloride (a salt) + water, and the general form acid + base gives salt + water. They note that with baking soda the products are salt + water + carbon dioxide. They use the data to explain the antacid: the base removes some H^+ from the stomach acid, so stomach pH rises from about 1.5 to 2 to about 3.5 to 5 and the burning feeling reduces. Learners answer: Why does the pharmacist not give a strong base like sodium hydroxide (pH 13) as an antacid? (safety, using the idea that sodium hydroxide is harmful, similar to soap but much stronger).	Say: 'Use only what we saw in the beaker and what we know about H^+ and OH^- to explain.' Ask each group to draw before and after pictures and then hold a gallery walk. Pose: 'Where did the H^+ ions go? What new substance formed?' WAIT TIME of 10-15 seconds, then cold-call 4 learners. Use Otieno's question about why lemon is sour to ask: 'If we add a base to lemon juice, what would happen to the sour taste and why?' Give the sentence frame: 'When ____ is added to ____, the ____ ions combine to form ____, so the pH ____.' Correct the misconception that the acid and base simply 'cancel out' by stating that H^+ and OH^- join to make water, and the remaining ions form a salt. Write the word equation and the ion equation H^+ (aq) + OH^- (aq) gives H_2O (l) on the board. Do not teach strong vs weak here; say that Lesson 8 will explore this.	Claim-Evidence-Reasoning (CER) with particle modelling. Learners state a claim (antacids relieve hyperacidity), cite evidence (beaker colour change and stomach pH data) and give reasoning with the particle model (H^+ + OH^- gives H_2O), which connects macroscopic observations to the particle level.	Collect each group's CER statement. Success indicators: the claim says the base neutralises the acid; the evidence cites at least one number (pH 1.5-2 rising to 3.5-5, or beaker colour change); the reasoning mentions H^+ and OH^- forming water. Check that the word equation is correct (magnesium chloride + water) and that learners do not say the acid 'disappears'. Ask 3 learners to give an exit explanation orally.
Driving Question Board (DQB) Creation  VIDEO: Physical and chemical changes <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Acids Bases and	Learners return to the DQB started in Lesson 1. They place sticky notes with answers to earlier questions (e.g. 'Why do patients with hyperacidity get antacids?' answered by neutralisation evidence) and add new questions such as 'Why does the stomach need acid at all?', 'Can too much antacid be harmful?', 'Is a strong acid neutralised the same way as a weak acid?' and 'How does lime help tea soil in Kericho?'. They sort questions into those answered, partly answered and still open.	Say: 'Look at the Driving Question Board. Which questions can we now answer with evidence from today?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to move a question to the 'answered' column and justify it with evidence. Invite every group to add at least one new question. Highlight links to upcoming lessons: strength and pH (Lesson 8) and uses and safety such as lime for Mr Kamau's soil in Kericho (Lesson 9). Ask: 'Does today's evidence help answer: does the alkalinity of a substance determine its use?'	Driving Question Board with sorting of questions. Sorting questions into answered, partly answered and open shows learners their own progress and shows the purpose of the next lessons, keeping the storyline coherent.	Check that each group has added at least one new, testable question and has used evidence from today's investigation to answer an older one. Look at whether questions move from 'what' to 'why' or 'how'. Note questions about strength and safety for Lesson 8 and Lesson 9 planning.

salts. Source: ARES Local Curriculum - Digital Literacy Content Search ARES for similar readings				
Model Building Phase  VIDEO: Neutralization Reaction - Overview Source: CK-12 Search ARES for similar videos  HTML: Acid-Base Neutralization (Flexbook) Source: CK-12 Search ARES for similar readings	Learners revise their particle-drawing model of H ⁺ and OH ⁻ ions in each product. Starting with lemon juice (pH 2.0), vinegar (pH 2.8), water (7.0), baking soda (8.3) and soap (10.0), they add a new panel: the stomach before antacid (many H ⁺ ions, pH about 1.5 to 2) and after magnesium hydroxide (fewer H ⁺ ions, pH about 3.5 to 5), with arrows showing H ⁺ + OH ⁻ forming H ₂ O. They write a short caption explaining neutralisation and one sentence on how this helps the patient in Nairobi.	Say: 'Take out your model from Lesson 6. What do we need to add or change after today?' Hand out colour pencils: one colour for H ⁺ , one for OH ⁻ , one for water. WAIT TIME of 10-15 seconds, then circulate asking: 'Where in your model do the OH ⁻ ions go when the antacid is added?' Cold-call 3 learners to share their revised panel and ask peers to add or challenge. Remind them that they will keep refining the model in Lessons 8 to 10.	Iterative model revision. Revising the same model across lessons lets learners see how new evidence (neutralisation) changes and extends the model, and builds toward the final explanation to the driving question.	Check models for: fewer H ⁺ ions after antacid; OH ⁻ ions shown reacting with H ⁺ to form H ₂ O; a salt shown or named; labels with pH values (1.5 to 2 before and 3.5 to 5 after). Common errors: showing H ⁺ ions disappearing without forming water, or showing pH falling after adding a base. Ask learners to fix these errors using the beaker data.

D. TEACHER REFLECTION

Did learners use H⁺ and OH⁻ ions to explain why the pH rose, rather than saying the acid and base 'cancel out'? Which groups misread the universal indicator colours? Were the predictions at the start different from the explanations at the end, and did learners notice the change in their own thinking? How well did learners connect the beaker to a real stomach and to Mwalimu's pharmacist story? What do I need to revisit in Lesson 8 about strength (for example, why a strong base is not used as an antacid) and in Lesson 9 about safe use of medicines?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When magnesium hydroxide (or baking soda) was added to dilute hydrochloric acid with universal indicator, the colour changed from red towards orange, yellow and green as the pH rose; adding water alone changed it very little. The stomach pH rises from about 1.5 to 2 to about 3.5 to 5 after an antacid.
What did I learn?	Neutralisation is a reaction between an acid and a base that gives salt + water. In terms of ions, H ⁺ (aq) + OH ⁻ (aq) gives H ₂ O (l). Antacids contain bases such as magnesium hydroxide or calcium carbonate.
How does this explain the phenomenon?	Patients with hyperacidity are given antacids because the base neutralises some of the excess hydrochloric acid in the stomach, forming salt and water and raising the pH to a safer level, so the use of a base as medicine depends on its alkalinity (and, as we will see, on its strength and safety).

LESSON 8 (80 minutes (2 x 40-minute periods)): Strength of Acids and Bases and the pH Scale

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners use universal indicator and the pH scale to compare strong and weak acids and bases, and to explain that strength depends on how much a substance ionises in water, which is different from how much is dissolved (concentration).
Knowledge	Learners will be able to: (1) use universal indicator colours and pH values to rank acids and bases by strength; (2) explain that a strong acid or base ionises almost completely in water while a weak one ionises only partly; (3) distinguish strength from concentration; (4) interpret that each pH unit is a 10-fold change in H^+ concentration, so lemon juice (pH 2.0) has about 6 times more H^+ (aq) than vinegar (pH 2.8).
Skills	Planning and carrying out investigations (NGSS SEP 3); analysing and interpreting data (SEP 4); using mathematics and computational thinking (10-fold change per pH unit) (SEP 5); developing and using models (SEP 2), drawing particle diagrams of H^+ and OH^- ions; practical skills in using universal indicator safely, observing, recording and comparing.
Attitudes	Learners appreciate that precise, evidence-based comparison (pH values rather than just 'sour' or 'slippery') is needed to decide how a substance is used safely. They show curiosity, careful teamwork and respect for laboratory safety rules.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life? (Focus today: how do we tell how strong an acid or base is, and why does strength matter for use and safety?)
Purpose in Storyline	Lessons 3-5 gave us indicators and the idea that acids release H^+ (aq) and alkalis release OH^- (aq). Lesson 7 showed antacids neutralising stomach hydrochloric acid. Lesson 8 now asks: do all acids behave the same? Learners find that acids and bases differ in strength, which refines the particle model (more or fewer ions) and prepares for Lesson 9 (uses and safety) and Lesson 10 (answering the driving question).
Safety Notes	Wear goggles and lab coats or aprons. Never taste any laboratory chemical. Use only 0.1 M solutions of hydrochloric acid, ethanoic acid, sodium hydroxide and ammonia, pre-prepared by the teacher. Sodium hydroxide is corrosive: handle with care and rinse spills with plenty of water. Ammonia has a pungent smell: keep tubes at arm's length, work in a ventilated room and do not sniff directly. Use dropper amounts only. Household samples (lemon juice, vinegar) may be used but must not be tasted in the lab. Wash hands after the practical.

B. LESSON OVERVIEW

Back in Mwalimu Achieng's classroom in Kisumu, the pH graph shows lemon juice at pH 2.0 and vinegar at pH 2.8, both acidic and both sour. Otieno asks: 'If both are acids, why is lemon juice lower on the graph?' Learners predict how acids and bases of the same concentration compare, then test 0.1 M hydrochloric acid, 0.1 M ethanoic acid (acetic acid), 0.1 M sodium hydroxide and 0.1 M ammonia solution with universal indicator. The data show HCl pH 1 versus ethanoic acid pH 3, and NaOH pH 13 versus ammonia pH 11. Learners explain the difference through ionisation, separate strength from concentration, calculate the 10-fold change per pH unit for lemon juice versus vinegar, link strength to antacid choice for hyperacidity (stomach hydrochloric acid, pH about 1.5 to 2), add questions to the Driving Question Board, and revise their particle model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Strong Acids and Bases - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Strong and Weak Acids Practice Source: CK-12  Search ARES for similar readings	Learners look again at the pH graph from the phenomenon (lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0). Individually they write a prediction in their notebooks: 'Two acids of the same concentration, 0.1 M hydrochloric acid and 0.1 M ethanoic acid, are tested with universal indicator. Will they have the same pH? Why or why not?' They do the same for 0.1 M sodium hydroxide and 0.1 M ammonia solution. They sketch a quick prediction on a pH strip from 0 to 14 showing where they think each will fall. Pairs compare predictions and note disagreements.	Display the pH graph and say: 'Lemon juice and vinegar are both acids and both sour, yet lemon juice is at pH 2.0 and vinegar at 2.8. Otieno asked why they are not the same. Today we find out what makes one acid stronger than another.' Give 3 minutes of silent writing. Say: 'Write your prediction and one reason. I am not marking right or wrong.' WAIT TIME 10-15 seconds after asking: 'Who thinks the two acids will have the same pH?' Take a show of hands, then cold-call 3 learners with different predictions: 'Wanjiru, what made you choose that?' Record the class split on the board (same / different) without judging. Ask: 'What could we measure to settle this?'	Prediction with reasoning (Predict-Observe-Explain): by committing to a prediction and a reason, learners expose their current ideas, such as 'same concentration means same acidity', so that the later evidence creates a productive conflict they must resolve.	Check notebooks for: (a) a clear prediction (same or different) for each pair; (b) a stated reason; (c) a pH strip sketch with the acids on the left of 7 and the bases on the right. Tally how many learners assume that equal concentration means equal pH. Look for the misconception 'strong means concentrated'. Note it for the Explain Phase.
Observe Phase  VIDEO: Strong acid solutions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Strong and Weak Bases Practice Source: CK-12  Search ARES for similar readings	In groups of 4, learners wearing goggles carry out a practical. They add 3 drops of universal indicator to 2 cm ³ of each pre-prepared 0.1 M solution in labelled test tubes: hydrochloric acid, ethanoic acid, sodium hydroxide and ammonia solution. They also test lemon juice and vinegar from the phenomenon. They match the colour to the universal indicator pH chart and record the pH in a table: HCl pH 1 (red), ethanoic acid pH 3 (orange), NaOH pH 13 (purple), ammonia pH 11 (blue-purple), lemon juice pH 2 (red-orange), vinegar pH 3 on the indicator chart (orange; pH meter value 2.8), and water pH 7 (green) as a reference. Each group plots the results on a class pH scale chart (0 to 14). They also observe that the HCl and ethanoic acid have the SAME concentration (0.1 M) but DIFFERENT pH, and likewise for	Before the practical, review the rules: 'Goggles on before anyone touches a tube. We never taste lab chemicals.' Demonstrate 3 drops, with the tube held at arm's length when testing ammonia. Circulate and ask each group: 'What colour do you see, and what pH does the chart give?' and 'Both tubes contain 0.1 M. What do you notice about the colours?' WAIT TIME 10-15 seconds before accepting an answer. Ask a group to read the data aloud, and cold-call 4 groups to report one result each to fill the class table. Provide pre-measured quantities and check that the sodium hydroxide is handled carefully. Remind: 'Compare to the pH chart in good light, against a white paper.' Reference the phenomenon: 'Does the pH of the lab acid match the pH of the lemon juice and vinegar we had in Lesson 1?'	Data table and number line (pH scale) construction: arranging all substances on one 0-14 scale makes patterns visible (strong acids near 0-1, weak acids near 3-6, weak bases near 8-11, strong bases near 13-14) and lets learners compare strength using evidence rather than taste.	Observe whether learners: (a) wear goggles and handle droppers correctly; (b) read the colour chart accurately (HCl pH 1, ethanoic acid pH 3, NaOH pH 13, ammonia pH 11); (c) record results in the correct units/format; (d) notice that equal concentration gave unequal pH. Use a checklist during circulation. Ask one probing question per group and note correct and incorrect answers.

	NaOH and ammonia.			
Explain Phase  VIDEO: Strong base solutions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Acid-Base Solutions Source: PhET Interactive Simulations (English)  Search ARES for similar readings	Learners analyse their data to answer: 'Why do 0.1 M HCl and 0.1 M ethanoic acid have different pH values?' The teacher introduces the idea of degree of ionisation with a particle drawing. In HCl nearly every molecule splits into H^+ (aq) and Cl^- (aq), so there are many H^+ ions and pH is 1. In ethanoic acid (acetic acid) only a few molecules ionise and most stay as whole molecules, so there are fewer H^+ ions and pH is 3. The same reasoning is applied to the bases: NaOH ionises almost completely into Na^+ and OH^- (aq) (pH 13), while ammonia produces fewer OH^- (aq) (pH 11). Learners complete a sentence frame: 'A strong acid _____ while a weak acid _____.' Then they separate strength from concentration: a dilute strong acid and a concentrated weak acid are discussed with a Kenyan example (a very watery HCl versus strong vinegar). Finally they use the pH graph: each pH unit is a 10-fold change in H^+ concentration, so lemon juice (pH 2.0, 1×10^{-2} mol/L) has about 6 times more H^+ than vinegar (pH 2.8, about 1.6×10^{-3} mol/L), since the difference of 0.8 pH units gives $10^{0.8} \approx 6.3$. They also compare soap (10.0) and baking soda (8.3), a difference of 1.7, and explain that soap is the stronger base in this comparison. Learners link this to Lesson 7: a pharmacist in Nairobi chooses an antacid such as magnesium hydroxide, a weak base, which neutralises excess stomach hydrochloric acid (pH about 1.5 to 2) and raises the pH to about 3.5 to 5 without making the stomach	Pose: 'Both tubes had 0.1 M. If concentration is the same, what else could cause a different pH?' WAIT TIME 10-15 seconds, then Think-Pair-Share for 2 minutes. Cold-call 3 pairs. Draw on the board two beakers: one with nearly all molecules split into H^+ and Cl^-, one with few molecules split. Say: 'Strength is about how much of the acid ionises, not how much is dissolved.' Use the analogy of 10 learners in a class: in a strong acid almost all 10 leave their seats (ionise), in a weak acid only 1 or 2 leave. Clarify language: 'Strong does not mean concentrated, and weak does not mean dilute.' Write on the board: pH 2.0 to 2.8 is a difference of 0.8, so $10^{0.8} \approx 6.3$, about 6 times. Ask: 'If one pH unit is 10 times, what would a difference of 2 units be?' (100 times) Cold-call 3 learners to explain. Address the misconception that Otieno raised: 'Lemon juice has more H^+ than vinegar, so it is more acidic, though both are weak acids.' Ask: 'Why do pharmacists in Nairobi prefer magnesium hydroxide, a weak base, to NaOH for hyperacidity?' Stress safety: NaOH is corrosive and must never be swallowed. Do not re-teach neutralisation itself (Lesson 7); use it as the application.	Claim-Evidence-Reasoning with particle modelling: learners claim that HCl is stronger than ethanoic acid, cite pH data as evidence, and use the ionisation model (more or fewer ions) as reasoning. Analogies and the 10-fold mathematical rule help them move from observation to a mechanism.	Collect exit-slip style responses: (a) 'Complete: HCl has a lower pH than ethanoic acid at the same concentration because _____. ' (b) 'True or false: a strong acid is always concentrated. Explain.' (c) 'Lemon juice (2.0) vs vinegar (2.8): how many times more H^+?' (about 6). Success indicators: uses the words ionise/ionisation, states that the number of H^+ (aq) ions is different, separates strength from concentration, and calculates correctly. Reteach in small groups if fewer than 70% give an acceptable answer.

	strongly basic like a strong base such as NaOH would.			
Driving Question Board (DQB) Creation  VIDEO: Strong Acids and Bases - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Identifying weak acids and strong acids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the DQB, which has the main question: 'Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications?' They review earlier questions (for example Otieno's 'Why is lemon sour?' and Wanjiru's 'Why is soap slippery?'). They mark questions that have been partly answered with evidence from this lesson, such as 'Why is lemon juice lower in pH than vinegar?' They write new questions on sticky notes, for example: 'Is a strong acid always more dangerous?', 'Why does soap, at pH 10, feel safe on the skin but NaOH, at pH 13, burns?', 'Why is the weak base in antacids better than a strong base?', 'How does the farmer in Kericho know how much lime to add?' and place them on the board under the headings 'Answered', 'Partly answered' and 'New'.	Say: 'Look at our DQB. Which questions can we now answer with evidence from today?' WAIT TIME 10-15 seconds. Cold-call 3 learners to move a question and give the evidence ('What data from today supports moving it?'). Then say: 'Write one new question that today's results made you wonder about, and link it to a product from our kitchen or a place in Kenya.' Invite 4 learners to read their questions aloud and group similar ones (strength and safety, strength and uses). Ask: 'Which of these will we investigate in Lesson 9 on uses and safety?' Mark the answered ones with the lesson number.	Question tracking and categorising: showing which questions have been answered, partly answered or newly raised keeps learners accountable for evidence and makes progress toward the driving question visible and student-owned.	Check the sticky notes: (a) Do new questions use strength, ionisation or pH ideas? (b) Are questions linked to the anchoring phenomenon? (c) Can learners justify moving a question by citing specific data (for example 'HCl pH 1, ethanoic acid pH 3')? Note learners who only restate old questions and prompt them with 'What did you see today that surprised you?'
Model Building Phase  VIDEO: Strong acid solutions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Identifying weak bases and strong bases Source: Khan Academy (English - US curriculum)  Search ARES	Learners revise their particle drawings of H ⁺ and OH ⁻ ions for each product from earlier lessons. On a new page they add a 'strength' section. For each of four beakers (0.1 M HCl, 0.1 M ethanoic acid, 0.1 M NaOH, 0.1 M ammonia) they draw the ions and the whole molecules that did not ionise, using more H ⁺ (aq) or OH ⁻ (aq) for strong substances and fewer for weak ones. They add a labelled pH scale (0 to 14) with the household products (lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0) and the lab solutions. They write a caption under lemon juice and vinegar: 'pH	Say: 'Revise your model so that someone from another class could tell which acid is stronger just by looking at your drawing.' Provide a model-checking list on the board: (1) are H ⁺ (aq) and OH ⁻ (aq) clearly labelled? (2) do strong substances show almost complete ionisation and weak ones only a few ions with many whole molecules? (3) does the pH scale include all products? (4) is the 10-fold rule shown? Note that H ₃ O ⁺ is simply another way of writing H ⁺ (aq) in water. Circulate and ask: 'Why did you draw fewer ions here?' WAIT TIME 10-15 seconds. Cold-call 3 learners to explain their	Cumulative model revision with peer critique: redrawing the particle model and comparing it with a partner's forces learners to turn the abstract idea of ionisation into a visible representation and to test it against the evidence (pH data).	Look at the models for: (a) correct ion counts (strong = many ions, weak = few ions plus intact molecules); (b) accurate pH placement of all substances; (c) correct use of H ⁺ (aq), OH ⁻ (aq); (d) the 10-fold rule applied to lemon juice and vinegar; (e) an explanation of strength distinct from concentration. Use a 3-level rubric (emerging, developing, secure) and record who needs support before Lesson 9.

for similar readings	2.0 has about 6 times more H ⁺ than pH 2.8.' They add a note explaining why a weak base such as magnesium hydroxide is used for hyperacidity. Learners compare models in pairs, then add one new feature and one question they still have.	model to the class. Close with: 'What does our model now say about lemon juice and vinegar? And what can it not yet explain?' Link to Lesson 9: 'Next we will use strength and safety to decide how acids and bases are used.'		
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D. TEACHER REFLECTION

Which learners still believe that 'strong' means 'concentrated'? Did the HCl versus ethanoic acid data (pH 1 versus pH 3 at 0.1 M) create enough conflict for them to question that idea? Were the particle drawings clear enough for learners to link pH to the number of ions? How well did learners manage the 10-fold calculation (lemon juice about 6 times more H⁺ than vinegar)? Was the practical safe and well organised, particularly the handling of sodium hydroxide and ammonia? What adjustments will help learners connect strength to safety and uses in Lesson 9?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	With universal indicator, 0.1 M hydrochloric acid showed pH 1 and 0.1 M ethanoic acid showed pH 3; 0.1 M sodium hydroxide showed pH 13 and 0.1 M ammonia showed pH 11. Lemon juice (pH 2.0) is lower than vinegar (pH 2.8) on the pH graph.
What did I learn?	Strength is about the degree of ionisation, not concentration. A strong acid or base ionises almost completely, while a weak one ionises only partly. Each pH unit is a 10-fold change in H ⁺ concentration, so lemon juice has about 6 times more H ⁺ (aq) than vinegar.
How does this explain the phenomenon?	At equal concentration, HCl gives many more H ⁺ (aq) ions than ethanoic acid, so its pH is lower; NaOH gives more OH ⁻ (aq) than ammonia, so its pH is higher. This helps explain why a weak base like magnesium hydroxide is a safer antacid for hyperacidity than a strong base, and it advances our answer that alkalinity, together with strength and safety, influences how products are used.

LESSON 9 (80 minutes (2 x 40-minute periods)): Uses of Acids and Bases and Safety: Why Vinegar Pickles, Soap Cleans, Lime Heals Soil and Antacids Soothe

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners evaluate how the chemical properties of acids and bases (pH, H ⁺ (aq) or OH ⁻ (aq) ions, reactions and strength) lead to their uses in food preservation, cleaning, baking, farming and medicine, and decide which safety precautions each use requires. This advances the driving question: Does the alkalinity of a substance determine its uses in day to day life?
Knowledge	By the end of the lesson learners should be able to: (1) link the acidic nature of vinegar (pH 2.8) and lemon juice (pH 2.0) to their uses in pickling and flavour; (2) link the basic nature of soap (pH 10.0) and baking soda (pH 8.3) to cleaning and baking mandazi; (3) explain how lime raises the pH of acidic soil in Kericho from about 4.5 to about 6.0 by neutralisation; (4) explain that antacids (magnesium hydroxide) are bases that neutralise excess stomach hydrochloric acid, raising stomach pH from about 1.5-2 to about 3.5-5; (5) state safety precautions that match the strength and nature of each substance.
Skills	Analysing and interpreting data (NGSS SEP 4); Constructing explanations (SEP 6); Engaging in argument from evidence (SEP 7); Evaluating and communicating information (SEP 8); Developing and using models (SEP 2); Critical thinking and decision making.
Attitudes	Learners appreciate the importance of acids and bases in everyday Kenyan life; show responsibility and care when handling household and laboratory chemicals; value respect for evidence and for each other's reasoning; show curiosity about how chemistry supports farming, health and food.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life?
Purpose in Storyline	Lessons 1-8 gave learners the classification of the four products (litmus, red cabbage, universal indicator), the Arrhenius ideas of H ⁺ (aq) and OH ⁻ (aq), reactions of acids with metals and carbonates, neutralisation and antacids, and strength versus concentration. Lesson 9 asks the 'so what?' question: do these properties actually decide how each product is used, and what precautions follow? It collects the use-evidence learners need for Lesson 10, where they will write the final evidence-based claim answering the driving question.
Safety Notes	Never taste any laboratory chemical or any household product used in the activity; the sour/slippy observations from Lesson 1 are discussed from the teacher's earlier demonstration only. Wear goggles and keep hands away from the eyes. Use only dilute, household-strength materials (vinegar, lemon juice, baking soda solution, soap solution, small amounts of garden lime in a closed bag). Lime (calcium hydroxide/calcium oxide) irritates skin and eyes: the teacher handles it and learners use only the pH data cards. Keep soap, lime and antacid tablets away from young children. Never mix household cleaners. Wash hands after the lesson and clean spills with plenty of water.

B. LESSON OVERVIEW

Mrs Achieng returns to the four products on display in the Kisumu classroom: lemon juice (pH 2.0), vinegar (pH 2.8), baking soda solution (pH 8.3) and soap solution (pH 10.0), with water (pH 7.0) as the reference line. Learners already know which are acids and which are bases, and why. Now they ask: do these properties decide how each product is used? In a Predict phase, learners guess which product would be best for four jobs: preserving kachumbari onions, cleaning greasy sufuria, raising mandazi and fixing acidic soil. In the Observe phase they rotate through four evidence stations (Mama mboga's pickling, soap cleaning, Mr Kamau's lime in Kericho, and the nurse/pharmacist in Nairobi giving antacids after spicy nyama choma) and record data in a use-property-safety table. In the Explain phase they build claim-evidence-reasoning statements linking property to use and then evaluate safety precautions, including why soap is harmful if swallowed even though it is useful. They add new questions and answers to the DQB, and revise their particle model of H⁺ and OH⁻ ions to show what each ion does in the use (for example OH⁻ (aq) from lime reacting

with H^+ (aq) in soil). Lesson 10 will then bring everything together into the final answer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: acids bases uses food HTML: Acid-Base Neutralization (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners look at the four products and the pH graph from Lesson 1 (lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0). Otieno asks, 'If lemon juice is so sour, can we use it to wash dishes?' and Wanjiru asks, 'Why is vinegar not used to make mandazi rise?' Learners get a prediction card with four jobs: (a) keeping kachumbari onions from spoiling, (b) removing grease from a sufuria, (c) making mandazi rise and soften, (d) helping maize grow in the acidic tea soil of Kericho (pH about 4.5), plus (e) relieving hyperacidity after nyama choma. Individually, for each job they write which product or type (acid or base) they would choose and why, then pair-share and draw a quick sketch of the ions (H^+ or OH^-) they think do the work.	Hold up each product and say, 'These all sit in the same kitchen, but would any one of them do every job? Pick the job, then pick the product. Write your reasoning, not just your answer.' Read the five jobs aloud slowly. Give WAIT TIME of 10-15 seconds of silent thinking before anyone writes. Circulate and note the ideas on a clipboard. Cold-call 4 learners (one per job) with, 'Tell us your prediction and one reason from our earlier lessons.' Do NOT confirm or correct. Say, 'Interesting. We will test that with evidence today.' Record 2-3 predictions in a 'Our predictions' column on the board, including any misconception such as 'all acids are dangerous and useless' or 'bases are always harmful'.	Prediction with reasoning (Think-Pair-Share): writing a prediction and a reason before instruction makes learners activate Lessons 1-8 (classification, ions, neutralisation, strength) and exposes misconceptions that the evidence stations can later confirm or challenge. Pair-sharing lets learners hear alternative ideas before committing.	Look for: (1) learners who link a use to acid or base nature (for example 'soap is a base, so it cleans'); (2) learners who mention neutralisation for soil or antacid; (3) learners who use pH numbers from the graph; (4) learners who give only taste or appearance as reasons (needs follow-up); (5) learners who mix up strength with concentration. Tally on a clipboard how many of the 5 jobs each learner matches to acid or base nature; target at least 3 of 5 with a reason. Learners who score below 3 receive a prompt during the Observe phase.
Observe Phase VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: acids bases uses food EXERCISE: Acid-Base Neutralization Practice <i>Related topic: not</i>	In groups of 4, learners rotate through four 8-minute evidence stations using data cards, photographs and safe demonstrations, and complete a Use-Property-Safety table. STATION 1 Pickling (acid): vinegar (5% ethanoic acid (acetic acid), pH 2.8) used by mama mboga for kachumbari and onions; a sealed jar of onions in vinegar vs onions in water (prepared two days earlier by the teacher) are compared for smell and appearance; learners record how the low pH and H^+ (aq) of vinegar make the environment	Before rotation say, 'Each station asks the same three questions: What is the use? Which property of the substance makes it work? What could go wrong if we are careless?' Assign roles (reader, recorder, reporter, safety officer). Set a timer for 8 minutes per station. Circulate and ask at least two groups per station: 'Where on the pH scale is this? Which ion, H^+ (aq) or OH^- (aq), is doing the work?' At Station 3 ask, 'What happens to the H^+ (aq) in the soil when lime is added? Where have we seen this before?'	Data-rich Gallery Rotation with a structured Use-Property-Safety table (Analysing and Interpreting Data, SEP 4): the same three questions at every station make learners search for a pattern across very different contexts (food, cleaning, farming, medicine), so they see that one idea (the acid or base nature, as shown by pH and ions) explains many uses. Comparing treated and untreated samples (onions, soil pH 4.5 vs 6.0, stomach pH 1.5-2 vs 3.5-5) supports cause-and-effect reasoning.	Observe whether each group's table (1) correctly labels each substance as acid or base using the pH data; (2) gives a property-based reason for the use instead of 'because it is useful'; (3) quotes numbers accurately (vinegar 2.8, soap 10.0, baking soda 8.3, soil 4.5 to about 6.0, stomach 1.5-2 to about 3.5-5); (4) gives a safety precaution that matches the substance. Use a 4-item checklist per group. Listen for the safety officer's checks. Common errors to note: saying lime 'makes the soil sweet' without reference to

<i>an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	unsuitable for many spoilage microbes, and note that lemon juice (pH 2.0) is used for flavour and tea. STATION 2 Cleaning and baking (bases): learners test greasy oil on a plate with water, vinegar, and soap solution (pH 10.0) and compare how well each lifts the grease; they also read the recipe card where Mama Wanjiru uses baking soda (pH 8.3) in mandazi and note that vinegar + baking soda produced CO₂ in Lesson 6, so baking soda releasing CO₂ helps dough rise. STATION 3 Farming: Mr Kamau's tea/maize farm in Kericho; a data card shows soil pH about 4.5 before lime and about 6.0 after liming, with a closed bag of lime shown by the teacher only. Learners use a soil pH strip card to place 4.5 and 6.0 on the pH scale and model the neutralisation of H⁺ (aq) in the soil by OH⁻ (aq) or carbonate from lime. STATION 4 Medicine: a card shows a patient in Nairobi with hyperacidity after eating spicy nyama choma; stomach acid is hydrochloric acid, pH about 1.5-2; a pharmacist gives a magnesium hydroxide antacid, raising the pH to about 3.5-5. Learners fill the table: use, nature (acid/base), property that explains the use, and safety precaution (for example vinegar: dilute, keep from eyes; soap: do not swallow, keep from children; lime: wear gloves and goggles; antacid: follow the dose, because too much gives side effects).	(neutralisation, Lesson 7). At Station 4 ask, 'How does the antacid change the stomach pH, and why does it not go to pH 7 or higher?' Allow WAIT TIME of 10-15 seconds after each question. Remind groups: 'No tasting anything. Safety officers, check goggles.' If a group is stuck, point to the pH graph instead of giving the answer.		neutralisation; saying all bases are safe to swallow because antacids are bases.
Explain Phase  VIDEO: <i>No closely matching video in the ARES library</i>	Each learner writes Claim-Evidence-Reasoning (CER) statements for two uses of their choice, for example: Claim: 'Vinegar is suitable for pickling	Model one CER aloud on the board using Mama Wanjiru's mandazi, then say, 'Now you do one about a job you predicted earlier. Which part of your claim	Claim-Evidence-Reasoning and Argument from Evidence (SEP 6 and 7): the CER frame forces learners to link a use (claim) to data (pH, ions, reactions) and to a	Collect CER statements as exit evidence and score with a 3-point rubric: Claim clear (1), evidence uses correct data (1), reasoning links to H⁺ (aq)/OH⁻ (aq),

for this activity.  Search ARES for videos Search terms: acids bases uses food  HTML: Acid-Base Neutralization (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	kachumbari.' Evidence: 'It has pH 2.8, lower than water (7.0), and contains H⁺ (aq) ions from ethanoic acid.' Reasoning: 'The acidic conditions help prevent spoilage, and vinegar is safe in food at 5% strength.' Then in groups they run a 'Safety Courtroom' on soap: Is soap a good product or a dangerous product? They argue for both, using that soap at pH 10.0 has OH⁻ (aq) of 1×10^{-4} mol/L, removes grease and is useful, but is harmful if swallowed, so the precaution is storage and use for washing only. Next, they compare strong and weak: Why is a weak acid (vinegar, ethanoic acid pH 3 at 0.1 M) safe in food but a strong acid (0.1 M HCl pH 1) is not, and why a weak base like magnesium hydroxide is chosen for antacids instead of a strong base such as sodium hydroxide (pH 13)? Finally, learners return to the driving question and propose: 'Does alkalinity determine use?' They conclude (with evidence) that nature, strength and safety together determine the use.	could a classmate challenge?' Give WAIT TIME of 10-15 seconds after each question. Ask for revisions of earlier predictions: 'Look at your prediction card. Which prediction would you keep, and which would you change? What evidence changed your mind?' Cold-call 5 learners (at least one per use: preservation, cleaning/baking, farming, medicine, safety) to share a claim and a piece of evidence. Press with: 'How do you know it is the pH and not just the look or the smell?' and 'Is strength the same as concentration? Explain with vinegar and HCl.' In the Safety Courtroom, give each side 3 minutes, then ask the whole class: 'Is a substance that is harmful if swallowed still useful? What precaution makes it safe to use?' Address misconceptions directly: 'Acids are not all dangerous (vinegar, lemon juice) and bases are not all safe (soap, lime, sodium hydroxide). Strength and concentration decide the risk.' Do not re-teach neutralisation or strength; point to the Lesson 7 and 8 posters.	chemical principle (reasoning), while the Safety Courtroom lets them weigh benefit against risk and see that strength matters. Explicit return to earlier predictions supports conceptual change.	neutralisation or strength (1). Target: 80% of learners score at least 2 of 3. Listen in the Safety Courtroom for correct use of 'weak vs strong' and 'concentration'. Watch for the misconception 'bases are safe because antacids are bases' or 'weak acid means dilute'; give a targeted follow-up question and revisit at the start of Lesson 10.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: acids bases uses food  EXERCISE:	Learners return to the Driving Question Board, which shows 'Does the alkalinity of a substance determine its uses in day to day life? Why are patients with hyperacidity issues given antacid medications?' They use sticky notes in two colours: green for questions they can now answer with evidence (for example 'Why is lime added to the soil in Kericho?' with the answer 'It neutralises acidic H⁺ (aq) and raises the pH from about 4.5 to about 6.0') and yellow for new or unanswered	Say, 'Look at our DQB. Which questions can we now answer with evidence? Which ones are still open?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to move a sticky note to the answered section and give the evidence in one sentence. Ask Otieno and Wanjiru (or the learners who first raised the questions) to read their original questions and say whether they have changed their answers. Ask, 'Which new question matters most for the driving question about	Driving Question Board as a living record of understanding: sorting questions into answered and open makes learning progress visible, connects each piece of evidence to the anchoring phenomenon, and gives learners ownership of what to investigate next.	Check that learners (1) move at least one question to 'answered' with a correct evidence statement; (2) write at least one new, testable question connected to the phenomenon (not 'what is an acid?'); (3) attach the right evidence card to the right question. Note the number of learners who contribute. Look for learners who can connect the antacid question to neutralisation and the soil question to pH change, and those who still have unanswered strength-versus-

Acid-Base Neutralization Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	questions (for example 'Why does a patient not take too many antacids?', 'Could we use lemon juice to clean?', 'Are all cleaners basic?'). They attach evidence cards (pickling table, soil pH, antacid pH change) to the relevant questions. In pairs they decide which questions are still open and which will be needed to answer the driving question in Lesson 10.	alkalinity and uses?' and have the class vote with a show of hands. Highlight any question about safety or strength for Lesson 10. Keep the DQB visible on the wall.		concentration questions.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: acids bases uses food  HTML: Acid-Base Neutralization (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners take out their particle drawing from Lesson 8 (H⁺ and OH⁻ ions in each product) and revise it to show uses. In groups they draw four mini-models: (1) vinegar in kachumbari with H⁺ (aq) ions present (low pH 2.8); (2) soap solution with OH⁻ (aq) ions (pH 10.0) around a grease particle; (3) Kericho soil with H⁺ (aq) before lime (pH about 4.5), and OH⁻ (aq) from lime combining with H⁺ (aq) to make H₂O after liming (pH about 6.0); (4) the stomach with H⁺ (aq) from hydrochloric acid (pH about 1.5-2) and OH⁻ (aq) from magnesium hydroxide forming water so fewer H⁺ (aq) remain (pH about 3.5-5). Each model has a key and a one-line caption explaining the use and a safety note. Learners then use a different colour pen to mark 'what changed in my model since Lesson 8' and write one sentence: 'The alkalinity of a substance helps determine its use because ...'.	Say, 'Take your Lesson 8 model. Today we add what the ions do in real life. Draw first, then label, then write a caption.' Circulate and check that ion counts reflect pH (fewer H⁺ at higher pH, more OH⁻ in bases). Ask, 'In the soil, where did the H⁺ (aq) go after lime? How do we show that in your drawing?' and 'Why does your stomach model still have some H⁺ (aq) left?' Give WAIT TIME of 10-15 seconds. Select 2 groups to display their models; cold-call 3 learners from other groups to say one thing that is strong and one suggestion. Note models that show H⁺ + OH⁻ giving H₂O. Do not give a finished model; ask learners to defend their drawing with data (pH values). Close with, 'Next lesson, you will use this model and all your evidence to answer the driving question.'	Developing and Using Models (SEP 2) with cumulative revision: updating the particle model each lesson makes the invisible ions visible and lets learners use the same model to explain very different uses (pickling, cleaning, soil, medicine), which strengthens the idea that one chemical principle explains many everyday observations.	Check each model for: (1) ions drawn in the right numbers or proportions for each pH; (2) correct neutralisation in soil and stomach (H⁺ + OH⁻ to H₂O); (3) a caption that links the ions to the use; (4) a safety note; (5) visible revision using a different colour. Use a 5-point checklist; target at least 4 of 5 for most groups. Photograph or collect a model from each group for Lesson 10. Learners who draw ions inconsistently with pH get a prompt at the start of Lesson 10.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Could learners connect each use to a specific property (pH, H⁺ (aq) or OH⁻ (aq), neutralisation) rather than just to the name of the product? (2) Which stations produced the strongest evidence-based talk, and which needed more scaffolding? (3) Did learners confuse strong/weak with concentrated/dilute when judging safety (for example vinegar vs 0.1 M HCl, magnesium hydroxide vs sodium hydroxide)? (4) Did every learner contribute to the Safety Courtroom and CER writing, including quieter learners? (5) Were the Kenyan contexts (Kisumu market lemons, mama mboga's kachumbari, Mama Wanjiru's mandazi, Mr Kamau's farm in

Kericho, the Nairobi pharmacist) meaningful and accessible to all learners? (6) Which DQB questions remain open, and what evidence will you need to bring to Lesson 10 so that learners can write a final claim about alkalinity, strength and uses? Record the learners who need extra support with the particle model before the consolidation lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Vinegar (pH 2.8) is used by mama mboga to pickle kachumbari and onions; soap solution (pH 10.0) lifts grease; baking soda (pH 8.3) is used for mandazi; lime raises Kericho soil from pH about 4.5 to about 6.0 for maize; a magnesium hydroxide antacid raises stomach pH from about 1.5-2 to about 3.5-5 after spicy nyama choma.
What did I learn?	The uses of acids and bases depend on their nature (H^+ (aq) or OH^- (aq)), their pH and their strength, and each use needs matching safety precautions: acids help preserve and flavour food, bases clean, bake and neutralise acids in soil and in the stomach, and not every acid or base is safe in the same way.
How does this explain the phenomenon?	Alkalinity largely determines uses because bases neutralise acids (acid + base gives salt + water): lime neutralises H^+ (aq) in acidic soil, antacids neutralise excess hydrochloric acid in hyperacidity, and soap's OH^- (aq) helps remove grease, while strength and safety decide how they can be used.

LESSON 10 (80 minutes (2 x 40-minute periods)): Consolidation: Answering the Driving Question – Alkalinity, Uses and Antacids

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate the whole unit by constructing an evidence-based final explanation that answers the driving question: does the alkalinity of a substance determine its uses in day to day life, and why are patients with hyperacidity given antacid medications?
Knowledge	Learners explain that acids (lemon juice pH 2.0, vinegar pH 2.8) release H^+ (aq) ions and bases/alkalis (baking soda pH 8.3, soap pH 10.0) neutralise acids or produce OH^- (aq) ions, and that these properties, together with strength and safety, determine uses such as pickling, cleaning, baking, liming soil in Kericho and treating hyperacidity. They explain that antacids neutralise stomach hydrochloric acid (acid + base gives salt + water), raising stomach pH from about 1.5 to 2 up to about 3.5 to 5.
Skills	Constructing explanations (NGSS SEP 6), engaging in argument from evidence (SEP 7), developing and using models (SEP 2), interpreting data (SEP 4), communicating scientific information (SEP 8).
Attitudes	Learners appreciate the role of chemistry in health, home and farming; show respect for peers' ideas when critiquing claims; and practise safety awareness when handling household acids and bases.
Key Inquiry Question	1. How do we classify substances as acids or bases? 2. Why are acids and bases important in our day-to-day life?
Purpose in Storyline	This is the culminating lesson. Learners return to the Kisumu kitchen phenomenon (lemon juice, vinegar, baking soda, soap and the pH graph) and use all evidence from Lessons 1-9 and their revised particle model to write and present the Final Explanation. They compare their Lesson 1 model with the final model and close out the Driving Question Board.
Safety Notes	No new practical chemicals are needed. If the pH graph is re-demonstrated with household products, never taste lab chemicals, wear goggles, wash hands after handling soap solution and keep lime and antacid samples away from the mouth and eyes. Emphasise that soap is harmful if swallowed and that antacids must be taken only as directed by a nurse or pharmacist.

B. LESSON OVERVIEW

Mrs Achieng's class in Kisumu returns to the opening display: lemon juice from Kisumu market, vinegar, baking soda and bar soap, with the pH graph (lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0). Learners first predict their final answer to the driving question, then review the evidence collected across the unit (indicator colours, H^+ and OH^- concentrations, reactions with metals and carbonates, neutralisation and stomach pH change, strong versus weak, and lime for Kericho soil). In the explain phase they write a Claim-Evidence-Reasoning Final Explanation, applying it to a nurse or pharmacist in Nairobi giving magnesium hydroxide to a patient with hyperacidity after spicy nyama choma. They then complete the Driving Question Board by marking questions answered, compare their Lesson 1 model with the final particle model of H^+ and OH^- ions in each product, and present to peers.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Neutralization	Learners look again at the pH graph and the four products from the Kisumu kitchen. Individually	Display the pH graph and the four products. Say: 'Eight weeks ago Otieno asked why lemon is sour	Think-Pair-Share with confidence rating: surfacing prior ideas and confidence lets learners compare	Check written predictions: look for use of the terms acid, base, pH, neutralise; note learners who

Reaction - Overview Source: CK-12 Search ARES for similar videos  HTML: Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	they write a 2-3 sentence prediction answering the driving question: 'Does the alkalinity of a substance determine its uses? Why are patients with hyperacidity given antacids?' They also rate their confidence from 1 to 3. Then they pair up (Think-Pair-Share) and compare answers, noting words they plan to use such as H⁺ (aq), OH⁻ (aq), pH and neutralisation.	and Wanjiru asked why soap is slippery. Today you will answer the big question completely. Write your best answer now, using what you know.' Give WAIT TIME of 10-15 seconds before pairs start talking after the question is posed, then allow 3 minutes of writing. Cold-call 4 learners to share one idea from their partner (not their own). Record key words on the board without judging. Ask: 'What evidence would convince a sceptical neighbour that your answer is right?'	their starting point with their later, evidence-based answer, which builds metacognition and shows how their thinking has grown.	connect antacids to stomach acid versus those who only say 'it cures pain'. Observable indicator: at least 4 of 6 pairs mention both pH/ions and a use. Note confidence ratings for later comparison.
Observe Phase  VIDEO: Neutralization Reaction - Overview Source: CK-12 Search ARES for similar videos  EXERCISE: Strong and Weak Acids Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	In groups of four, learners complete an Evidence Gallery Walk with six stations made from their own notebooks and class charts: (1) pH graph data: lemon juice 2.0, vinegar 2.8, water 7.0, baking soda 8.3, soap 10.0 and the 10-fold H⁺ change per pH unit (lemon juice about 6 times more H⁺ than vinegar); (2) indicator colours: litmus, red cabbage and universal indicator; (3) H⁺ and OH⁻ concentrations: lemon juice 1 x 10⁻² mol/L, vinegar about 1.6 x 10⁻³ mol/L, soap OH⁻ 1 x 10⁻⁴ mol/L; (4) reactions: vinegar + baking soda fizzing with CO₂ turning limewater milky, Mg + HCl giving H₂ with a squeaky pop; (5) neutralisation and stomach pH: about 1.5 to 2 rising to about 3.5 to 5 after an antacid of magnesium hydroxide; (6) uses and strength: vinegar for pickling kachumbari, soap for cleaning, baking soda for Mama Wanjiru's mandazi, lime in Kericho raising soil pH from about 4.5 to 6.0 for maize, and HCl pH 1 versus ethanoic acid (acetic acid) pH 3. At each station they record	Set a timer for 4 minutes per station. Circulate and ask: 'Which number on this chart best supports a use? Why that number?' and 'Is this about strength or about concentration?' At Station 6 prompt: 'Mr Kamau in Kericho adds lime. What is the lime doing to the H⁺ ions?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to state the strongest piece of evidence they have found so far. Correct misconceptions promptly, for example that all bases are alkalis (only those that dissolve in water giving OH⁻ (aq) are alkalis) or that a strong acid means a concentrated acid.	Evidence Gallery Walk with Evidence Cards: collecting data from several sources into one set of cards lets learners see patterns across different kinds of evidence and prepares them to select the most relevant data for a claim.	Check each group has at least six Evidence Cards with correct values and the lesson source. Observable indicators: learners quote exact pH values, distinguish strength from concentration when questioned, and link at least one piece of evidence to each of the following: classification, ions, reactions, neutralisation, uses.

	one piece of evidence on an Evidence Card and note which lesson it came from.			
Explain Phase  VIDEO: Evaluating a source's reasoning and evidence <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Individually, learners write their Final Explanation using the Claim-Evidence-Reasoning (CER) frame. Claim: answers the driving question, that alkalinity, together with strength and safety, largely determines uses. Evidence: at least four data points, for example soap pH 10.0 and its slippery feel for cleaning, vinegar pH 2.8 for pickling, lime raising Kericho soil from about 4.5 to 6.0, and stomach pH rising from about 1.5 to 2 to about 3.5 to 5 after an antacid. Reasoning: link to ions, for example acids release H^+ (aq), bases/alkalis neutralise acids or give OH^- (aq), and H^+ (aq) + OH^- (aq) gives H_2O, so acid + base gives salt + water. They must specifically explain why a patient in Nairobi with hyperacidity after eating spicy nyama choma is given magnesium hydroxide: excess stomach hydrochloric acid is neutralised, forming salt and water and raising stomach pH to a safer level. They also address the limits: strong bases such as sodium hydroxide (pH 13) are unsafe to swallow, which is why weak bases are chosen for antacids. Then in pairs they critique each other's CER using a checklist and revise.	Model one sentence stem only: 'I claim ____ because the data show ____, which means ____.' Say: 'Do not just state a fact. Tell me which number proves it and what the ions are doing.' Give 8 minutes of silent writing. Circulate and ask: 'Why is soap pH 10.0 good for cleaning but not for drinking?' and 'Why not give the patient sodium hydroxide at pH 13?' Allow WAIT TIME of 10-15 seconds. Then run a peer critique: pairs swap and tick the checklist (claim, 4 data points with values, ion reasoning, antacid reaction, safety). Cold-call 5 learners to read one strong reasoning sentence aloud, and invite the class to say whether it uses evidence.	Claim-Evidence-Reasoning (CER) with peer critique: structuring the argument forces learners to connect data to a scientific principle, and peer critique exposes gaps so explanations become more complete and accurate.	Use the CER checklist: (1) claim answers both parts of the driving question; (2) at least four correct values cited; (3) reasoning refers to H^+ (aq) and OH^- (aq) and neutralisation; (4) antacid explained as acid + base gives salt + water with stomach pH change; (5) mention of strength or safety. Observable indicator: at least 80% of learners meet 4 of 5 criteria; those below receive a targeted sentence-stem prompt.
Driving Question Board (DQB) Creation  VIDEO: Using a pKa table <i>Related topic: not an exact match for this lesson.</i>	The whole class revisits the Driving Question Board started in Lesson 1. Groups take the sticky notes, including Otieno's 'Why is lemon sour?' and Wanjiru's 'Why is soap slippery?', and sort them into 'Answered', 'Partly answered' and 'Still wondering'. For each answered question, a learner	Say: 'Walk to the board. Pick a question that you can now answer with evidence, not just opinion.' Allow WAIT TIME of 10-15 seconds before groups move. Ask 6 learners (cold-call) to read a question and state the evidence that answers it. Probe: 'Which lesson gave us that? What would	Driving Question Board review: sorting questions by how well they have been answered makes the growth in understanding visible and gives learners ownership of the learning path, linking each question to specific evidence.	Observe whether sorted questions are matched to correct evidence (for example, soap slipperiness to OH^- (aq) from alkalis, sourness to H^+ (aq)). Observable indicators: every group attaches at least two evidence cards with lesson numbers; at least one thoughtful new question is written per group.

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Strong and Weak Acids Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	attaches the evidence card and the lesson number. Learners write one new question for future study, such as why bee stings or ant bites feel painful or how antacid tablets differ in strength, and add it to a 'Next Questions' area.	we still need to test?' Highlight the unit's two key inquiry questions: how we classify acids and bases, and why they are important in daily life. Accept honest 'partly answered' notes, for example that learners have not yet measured exact stomach pH themselves.		
Model Building Phase VIDEO: Data to justify experimental claims examples <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Strong and Weak Acids and Acid Ionization Constant (Ka) (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners take out their Lesson 1 model (their first idea of why the four products differ) and draw their final particle model on a new page: four beakers (lemon juice, vinegar, baking soda solution, soap solution) showing H⁺ (aq) and OH⁻ (aq) ions, with more H⁺ in lemon juice (pH 2.0) than vinegar (pH 2.8), more OH⁻ in soap (pH 10.0) than baking soda (pH 8.3), and water at pH 7.0 as the neutral reference. They add a fifth drawing of the stomach with H⁺ (aq) and Cl⁻ (aq) from hydrochloric acid, an antacid of magnesium hydroxide adding OH⁻ (aq), and H⁺ + OH⁻ forming H₂O, with a pH arrow from about 1.5 to 2 up to about 3.5 to 5. Using a two-column table they compare 'Lesson 1 model' and 'Final model', listing what changed, and then present their final model to a partner and then to the class. They also fill in the summary table of what they observed, learned and explained.	Say: 'Put your first model next to your new one. What have you added that you could not explain before?' Allow WAIT TIME of 10-15 seconds, then 6 minutes for drawing. Check that ion numbers match the pH ordering (lemon juice < vinegar < water < baking soda < soap). Ask: 'Where do the H⁺ ions go when the antacid is swallowed?' and 'Why does the stomach pH rise but not reach 7?' Cold-call 4 learners to present one change between models. Use the Mama Wanjiru mandazi example to ask where the baking soda fits in the model. Remind learners that H₃O⁺ is the form H⁺ takes in water, mentioned only as a note.	Model comparison (before and after): placing the first and final models side by side makes conceptual change explicit, helping learners articulate how the particle-level explanation accounts for the phenomenon and its everyday uses.	Rubric for the final model: (1) H⁺ and OH⁻ ions correctly placed in each product; (2) relative amounts match the pH graph; (3) neutralisation in the stomach shown as H⁺ + OH⁻ giving H₂O; (4) pH arrow before and after the antacid; (5) at least two written changes from Lesson 1. Observable indicator: learners can explain, without notes, why the antacid helps a hyperacidity patient and why lime helps Kericho soil.

D. TEACHER REFLECTION

After the lesson, review the CER write-ups and final models. Which learners still confuse strength with concentration, or treat all bases as alkalis? Did learners use

specific pH values and ion reasoning rather than general statements? Did the antacid explanation include both the neutralisation reaction and the stomach pH change? Note which stations of the gallery walk prompted the best discussion and which evidence was ignored. Consider how participation differed across groups, whether any learners needed language support in writing, and what follow-up (for example, a short quiz or a home task interviewing a local pharmacist or farmer about antacids or lime) would consolidate learning.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that lemon juice (pH 2.0) and vinegar (pH 2.8) are acidic, sour and turn blue litmus red, while baking soda (pH 8.3) and soap (pH 10.0) are basic, bitter or slippery and turn red litmus blue. We also observed that acid + base reactions give salt + water, that stomach pH rises from about 1.5 to 2 to about 3.5 to 5 with an antacid, and that lime raises Kericho soil pH from about 4.5 to 6.0.
What did I learn?	We learned that acids release H^+ (aq) ions and alkalis produce OH^- (aq) ions, that lower pH means more H^+ , that strength (degree of ionisation) differs from concentration, and that properties such as pH, strength and safety determine how acids and bases are used for pickling, cleaning, baking, farming and medicine.
How does this explain the phenomenon?	We explain that alkalinity, together with strength and safety, largely determines the uses of substances, and that patients with hyperacidity are given antacids such as magnesium hydroxide because the base neutralises excess stomach hydrochloric acid (H^+ (aq) + OH^- (aq) gives H_2O , acid + base gives salt + water), raising the stomach pH to a safer level.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.